

# Stylized Facts on Multinational Firms and Trade in Latin America

Each fact with its original exhibit reproduced on the current base and the variants for the working paper — all goods

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# 1 How to read this document

The document follows the six stylized facts of the July-2026 note (v. 2026-07-29). Each section opens with the fact, reproduces its *original* exhibit on the current base, and then places under it every variant prepared for the working paper: the figures with the foreign-MNE bar split by the multinational’s home country (item 1a); alternative product-sophistication measures for Figure 2, including the Fontagné–Guimbard–Orefice (2022) elasticities, with the full distributions (1b); two-way tables and heat maps of parent country  $\times$  destination (1c, 1d); the decomposition of multinationals into foreign vs domestic and of foreign MNEs into present-through-headquarters vs not, in the presence and distance regressions (1e); products at HS6 with descriptions (1f). Revision 6 (2026-09-18) adds the destination side of Fact 3—who carries the exports that reach the United States, China and the EU-27, and which lines US-parent MNEs ship home—and the cross-sector summaries (US-parent share by origin  $\times$  sector, home share by parent  $\times$  sector, Facts 5 and 6 in one table each). The four sectors are summarised at the end; the sector-by-sector version of every exhibit is the companion document *WP\_sectors*, and the country-by-country exhibits are in *WP\_countries*. Everything is generated by `stylized_facts/python/wp_*.py` in the project repository ([https://github.com/jsvp97/IADB\\_Orbis\\_DNB\\_Customs\\_Final](https://github.com/jsvp97/IADB_Orbis_DNB_Customs_Final)); the reader’s guide is `docs/WP_RESULTS_2026-09-07.md`.

**Conventions (identical to the note).** Matched = the firm was found in Orbis or D&B (`_merge_DNB_Orbis=3`); domestic MNE = parent in the exporting country; foreign MNE = matched minus domestic, so matched firms with no recorded parent count as foreign; nine origins (Ecuador excluded); value-weighted; pooled 2006–2022; navy = foreign, light gray = domestic. Wherever the foreign bar is split by home country, the ten largest parent countries (by foreign-MNE export value in the scope) are shown in shades of navy (rank 1 darkest) and every other foreign MNE—including those with no recorded parent—is pooled into “Other foreign MNEs” (gray-blue); domestic MNEs keep the light gray. Two-way country tables show the top-10 parents and destinations; heat maps the top 15. Regressions use plain logs (a zero count drops the cell from that column; no  $\ln(1+x)$ ) and cluster at origin–destination. Sectors: agriculture = HS 01–24; mining & fuels = HS 25–27 and 71; manufacturing = HS 28–97 excluding 71; rest = HS 98–99, chapter 00 and unclassifiable codes (services are not in customs data).

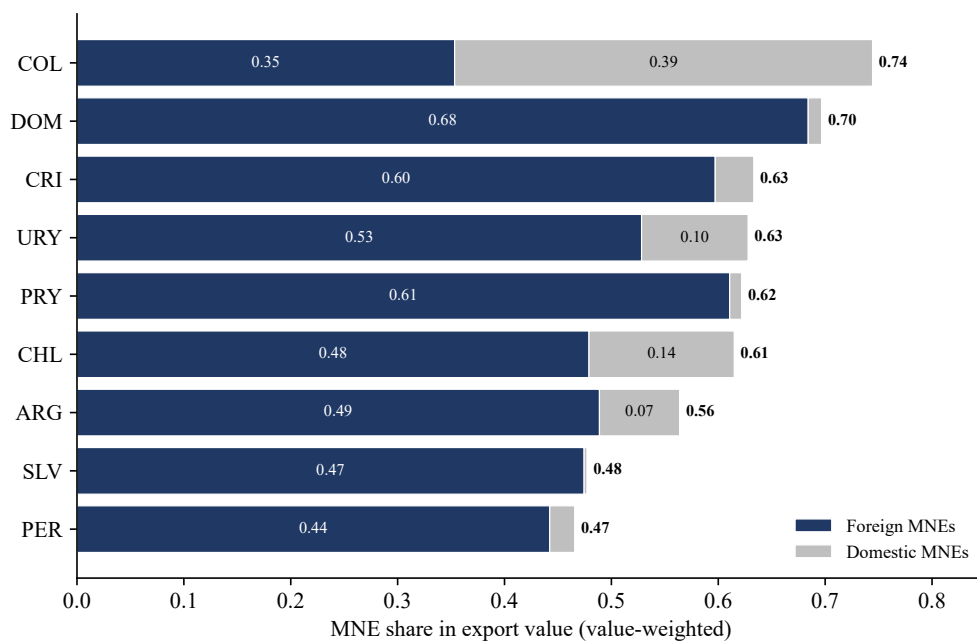
**Same data as the note.** The regressions that depend only on the matched flag reproduce the note to the last digit: Table 27 returns the note’s Table 1 Panel A coefficients and every observation count (2.0823 / 2.0685 / 0.9414; extensive 1.4622 / 1.4591 / 0.6818;  $N = 1,008,472 \dots 1,137,550$ ) and Table 32 the note’s Table 2 pooled coefficients (−0.1639 / −0.1848; 0.0461 / 0.0458;  $N = 5,081,547 / 5,081,360$ ); Figure 24 gives the note’s 56 % of value for parents with more than 100 affiliates. The cells that use the group’s affiliate links differ in the third decimal only (Table 1 Panel B 1.3536 vs the note’s 1.3522; Table 2 “present” 0.0689 vs 0.0673), because the current base carries the affiliate and parent links of the 2026-04-21 build. The base is therefore the same 5.45 M firm–destination–product–year rows. The field that differs materially is the *parent country*: the current base carries a recorded parent for 92 % of foreign-MNE export value (script 02 and the AI review of unknown parents, merged in the 2026-04-21 build), whereas the note’s Figure 4 was drawn on a copy in which about half of that value had no recorded parent. The parent ranking becomes USA > GBR > CAN > LIE > DEU > NLD (Liechtenstein: foundations owning Argentine and Chilean exporters of copper ore, oil-cake and cereals—a conduit), and the domestic-MNE share rises where the recovered parents are the origin country (COL 0.28  $\rightarrow$  0.39, CHL 0.02  $\rightarrow$  0.14), so Fact 1’s “foreign  $\gg$  domestic” needs a qualification for Colombia.

**Two data limits.** (i) “Headquarters exporters” are not identifiable: every matched exporter has a global ultimate owner different from itself (0 of 5.45 M rows), because the match runs through ownership links. What is identifiable is the group’s presence *at the destination* through its headquarters (the affiliate ships to the parent’s country); Facts 5 and 6 decompose foreign MNEs on that. (ii) On the substitution elasticities: Fontagné, Guimbard and Orefice (2022) estimate one import-demand elasticity per HS6 product (rev. 2007), common to all exporters and importers—the elasticity is identified from tariff variation across the universe

of bilateral flows of the product and is a characteristic of the product, not of the origin. Assigning the same  $|\sigma|$  to a product whatever its exporting country is therefore the intended use of the database (non-significant or positive HS6 estimates are replaced by the source with the HS4 average and flagged); no origin-specific adjustment is needed or possible. The Rauch (1999) classification (conservative, SITC 3/4 digits mapped to HS6) is added; the Micro-D classification of Bernini, González, Hallak and Vicondoa (2018) is defined on Argentina's 12-digit nomenclature from package-size attributes and cannot be carried to HS6 data for other countries.

## 2 Fact 1 — Multinational corporations account for a large share of export values across countries

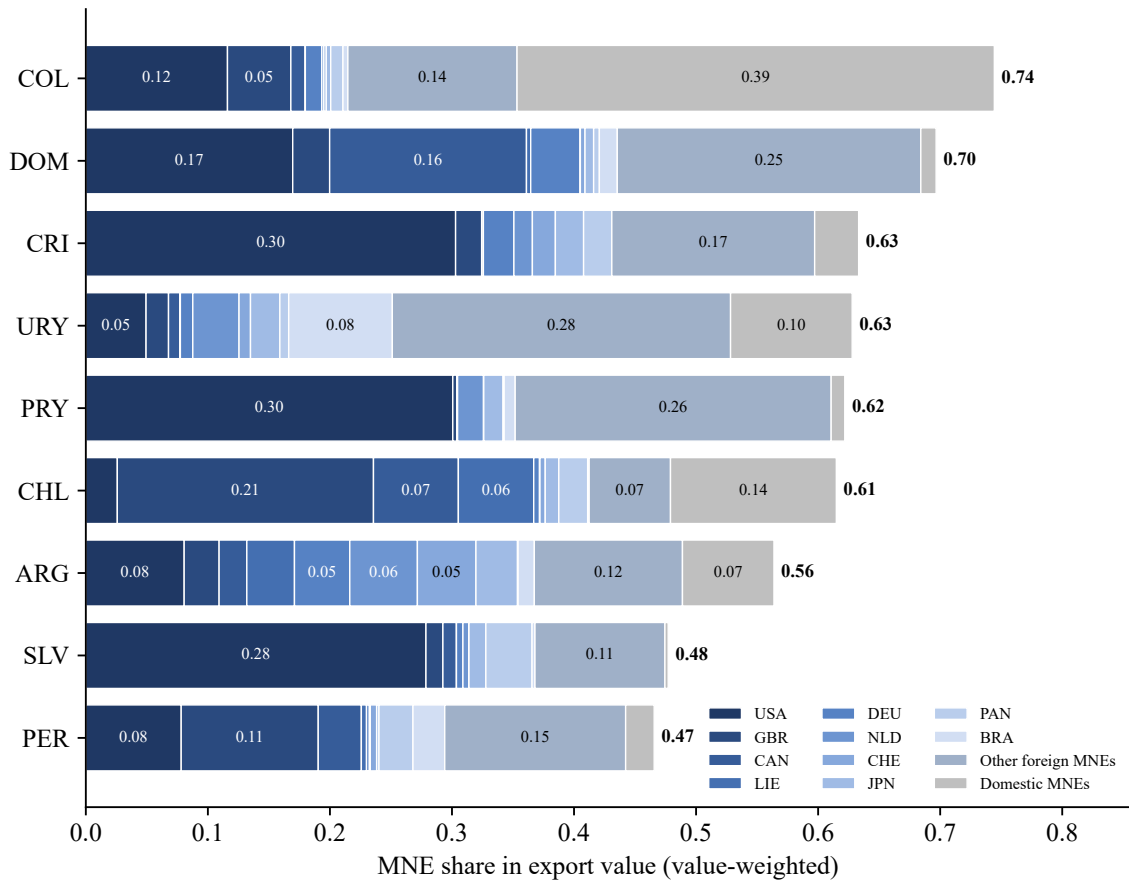
### 2.1 Original exhibit



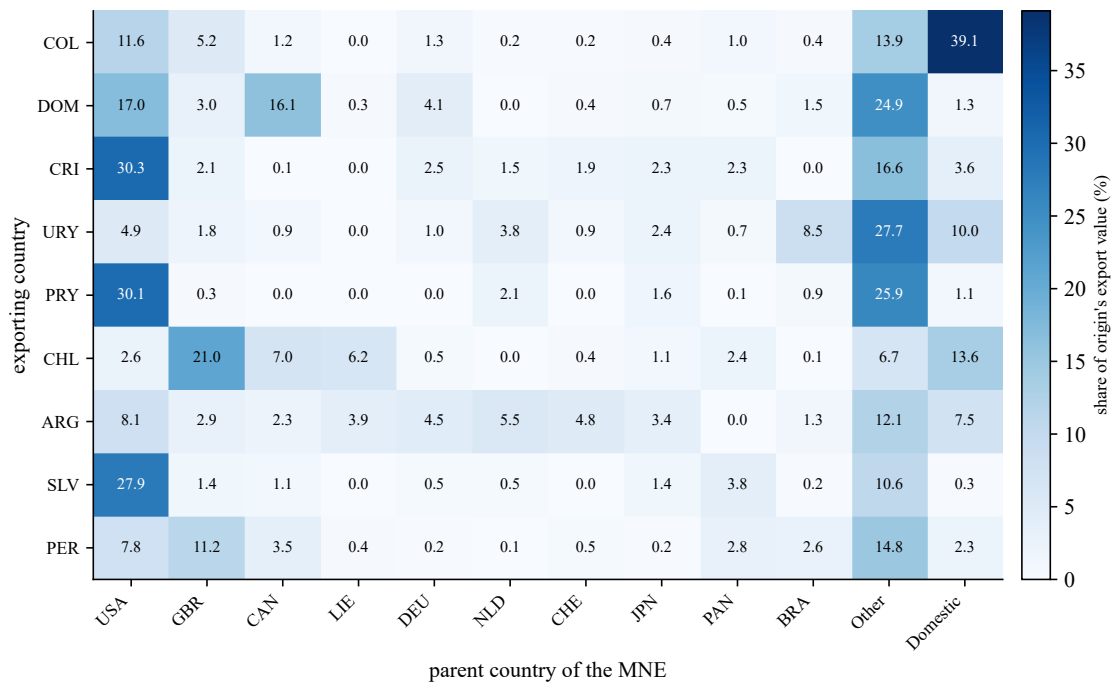
**Figure 1:** Figure 1 of the note, reproduced on the current base: MNE share of export value by origin, foreign vs domestic.

*Note:* Value-weighted; pooled 2006–2022; nine LAC origins. Bar length = total MNE share; numbers in the appendix.

## 2.2 Variant: the foreign bar split by the multinational's home country (item 1a)

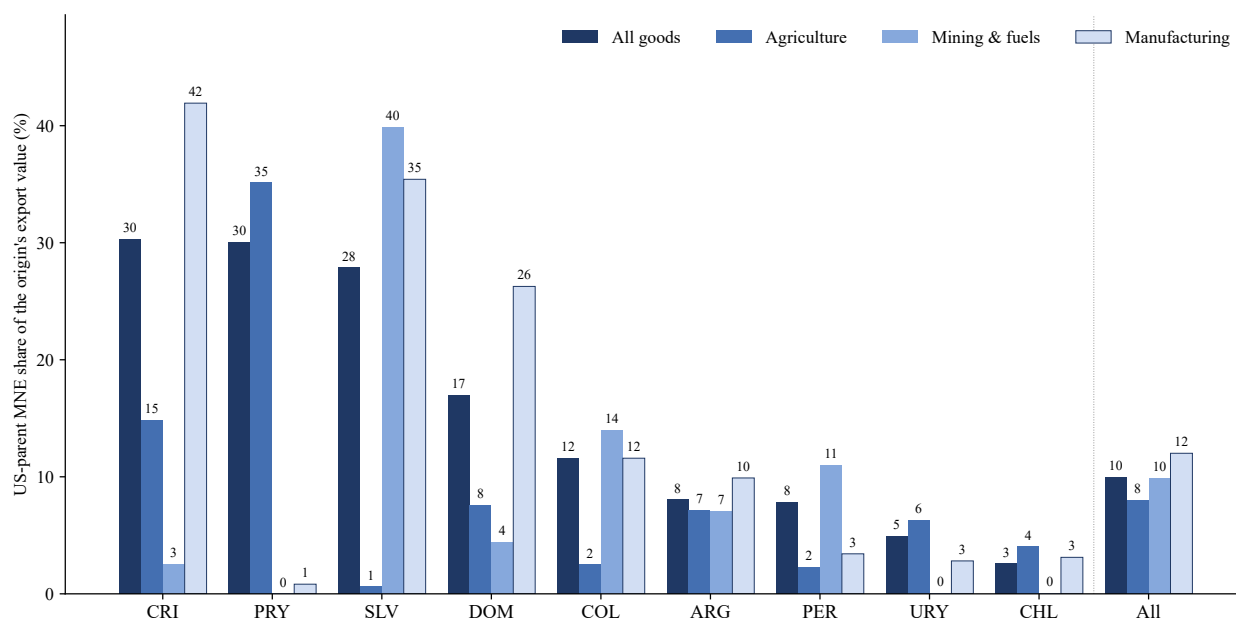


**Figure 2:** MNE share of export value by origin, foreign bar split by parent country (top 10, Other foreign, Domestic).



**Figure 3:** companion heat map: share of each origin's export value by parent-country group (%).

## 2.3 Variant: the US-parent share of exports by origin and sector (revision 6)



**Figure 4:** Share of each origin's export value moved by multinationals with a US parent, all goods and by sector.

*Note:* Share of the cell's export value (all destinations); origins sorted by the all-goods share. Sector versions of Figure 2 are in *WP\_sectors*; this figure replaces them for the US-parent question.

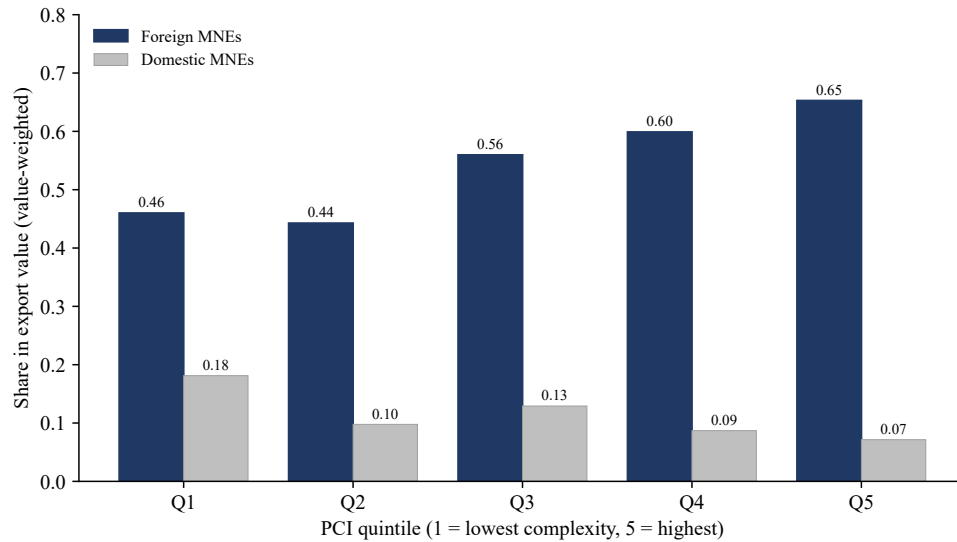
**Table 1:** US-parent MNE share of exports and US-parent export value, by origin and sector

| Origin | US-parent MNE share of the cell's exports (%) |             |                |               | US-parent MNE exports (\$bn) |             |                |               |
|--------|-----------------------------------------------|-------------|----------------|---------------|------------------------------|-------------|----------------|---------------|
|        | All goods                                     | Agriculture | Mining & fuels | Manufacturing | All goods                    | Agriculture | Mining & fuels | Manufacturing |
| CRI    | 30.3                                          | 14.9        | 2.5            | 41.9          | 31.7                         | 6.5         | 0.0            | 25.2          |
| PRY    | 30.1                                          | 35.1        | 0.0            | 0.8           | 19.7                         | 19.6        | 0.0            | 0.1           |
| SLV    | 27.9                                          | 0.6         | 39.9           | 35.4          | 18.2                         | 0.1         | 0.8            | 17.3          |
| DOM    | 17.0                                          | 7.6         | 4.4            | 26.3          | 11.7                         | 1.2         | 0.7            | 9.8           |
| COL    | 11.6                                          | 2.5         | 14.0           | 11.6          | 61.3                         | 2.1         | 46.7           | 12.4          |
| ARG    | 8.1                                           | 7.2         | 7.1            | 9.9           | 48.2                         | 24.6        | 4.2            | 19.4          |
| PER    | 7.8                                           | 2.3         | 11.0           | 3.4           | 32.4                         | 1.8         | 27.8           | 2.8           |
| URY    | 4.9                                           | 6.3         | 0.0            | 2.8           | 3.5                          | 2.8         | 0.0            | 0.7           |
| CHL    | 2.6                                           | 4.0         | 0.1            | 3.1           | 12.3                         | 4.2         | 0.1            | 7.0           |
| All    | 10.0                                          | 8.0         | 9.9            | 12.0          | 238.9                        | 62.9        | 80.3           | 94.7          |

Share of each origin's export value (all destinations) moved by multinationals whose recorded ultimate parent is in the United States, all goods and by sector; origins sorted by the all-goods share. Sectors as in the conventions (rest not shown, included in all goods). Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched.

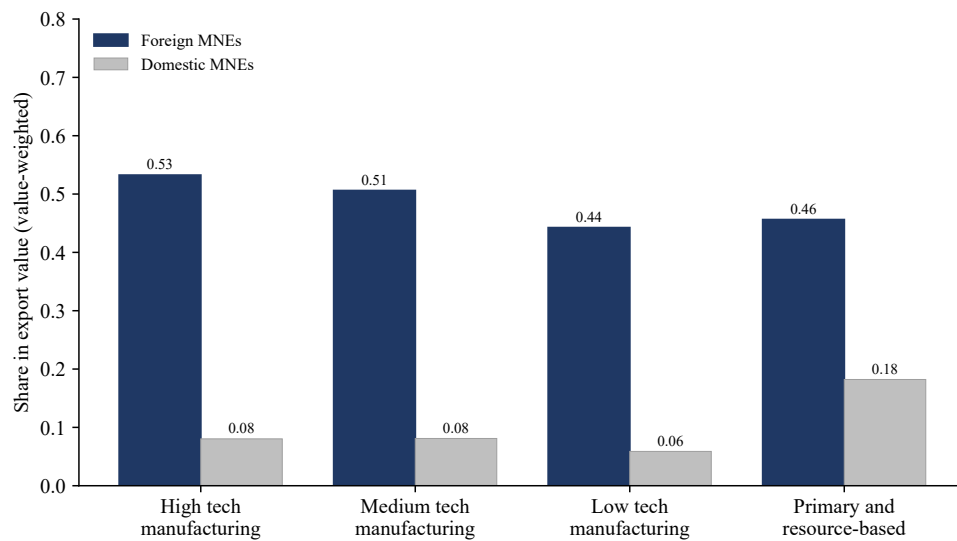
### 3 Fact 2 — Foreign multinationals specialize in complex products, domestic ones in primary goods

#### 3.1 Original exhibits



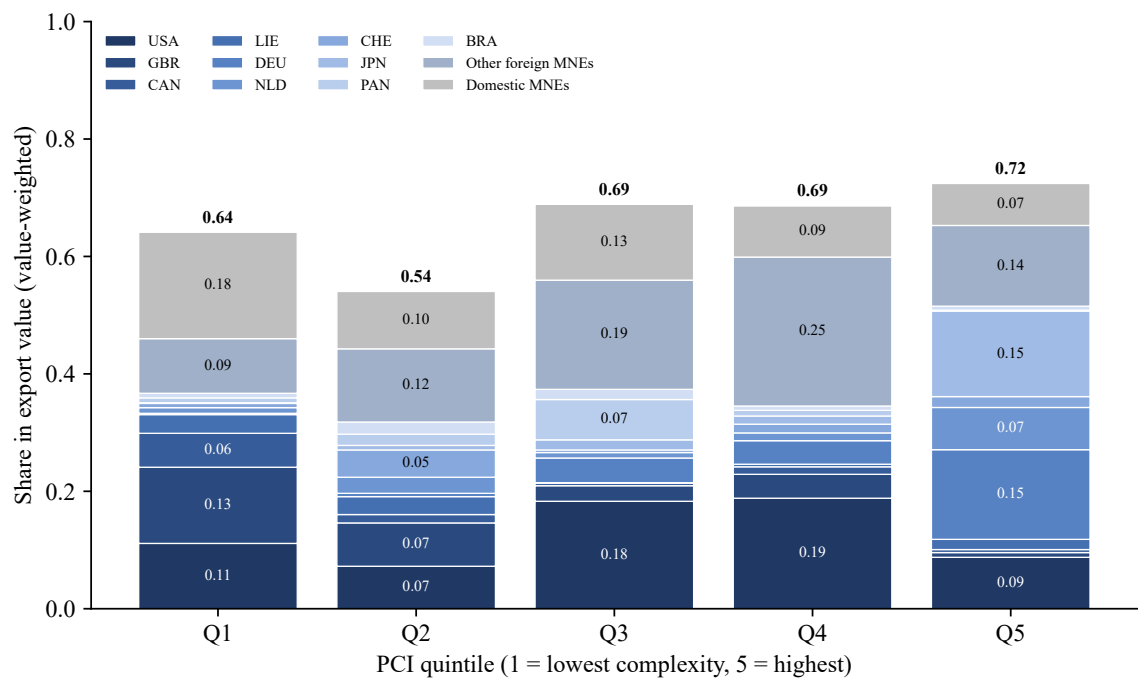
**Figure 5:** Figure 2 of the note, reproduced: MNE share of export value by Product Complexity Index quintile.

*Note:* Value-weighted at HS6; quintiles of the Hausmann–Hidalgo PCI over the scope’s HS6 products (Q1 = least complex).

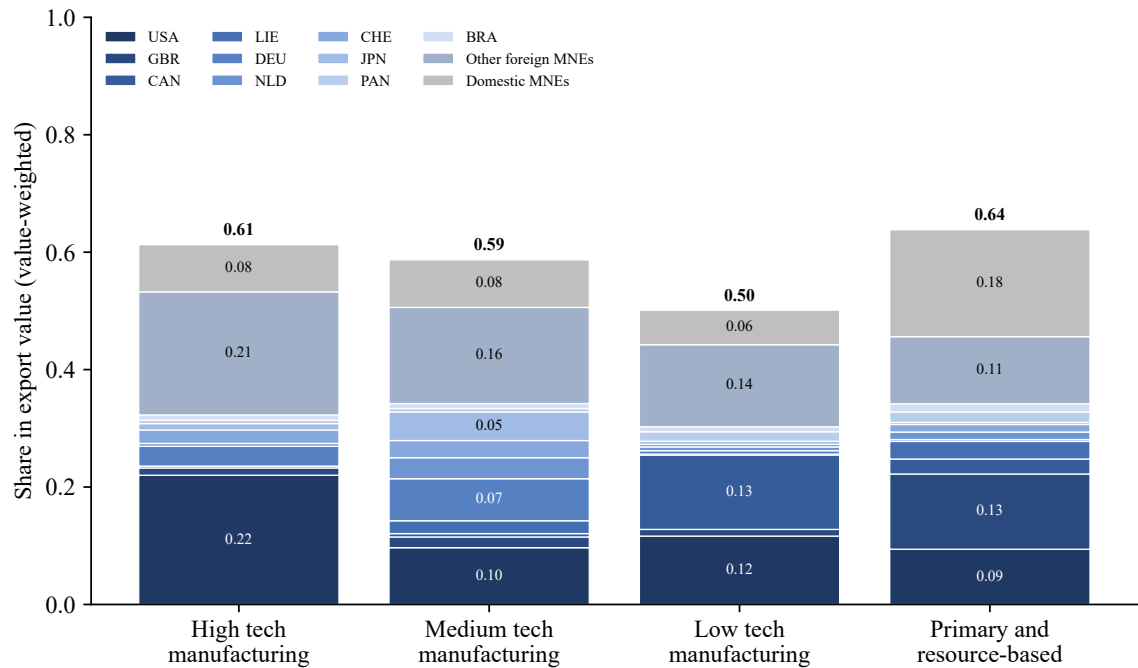


**Figure 6:** Figure 3 of the note, reproduced: MNE share of export value by Lall (2000) technology category.

### 3.2 Variant: split by the multinational's home country (item 1a)

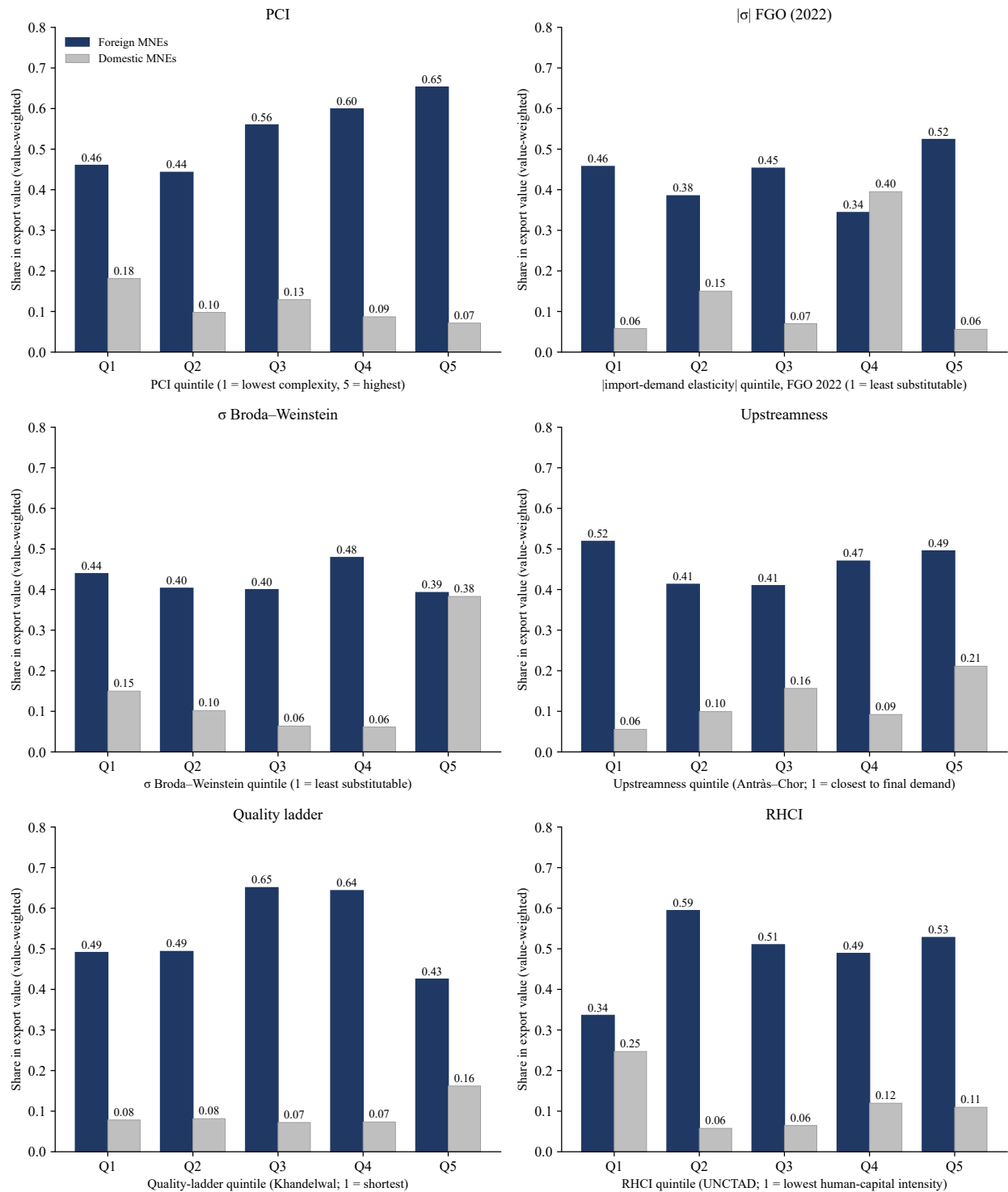


**Figure 7:** Figure 2 with the foreign bar split by parent country.

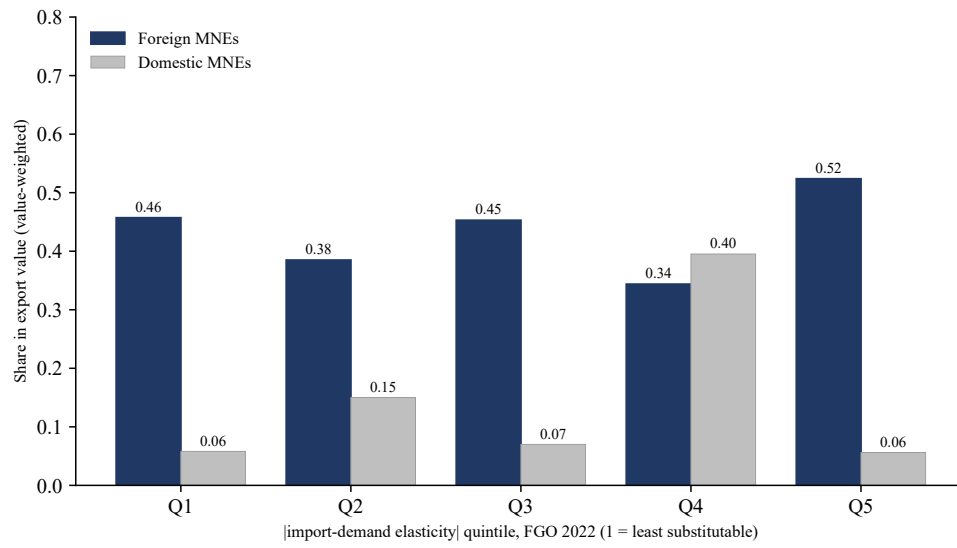


**Figure 8:** Figure 3 with the foreign bar split by parent country.

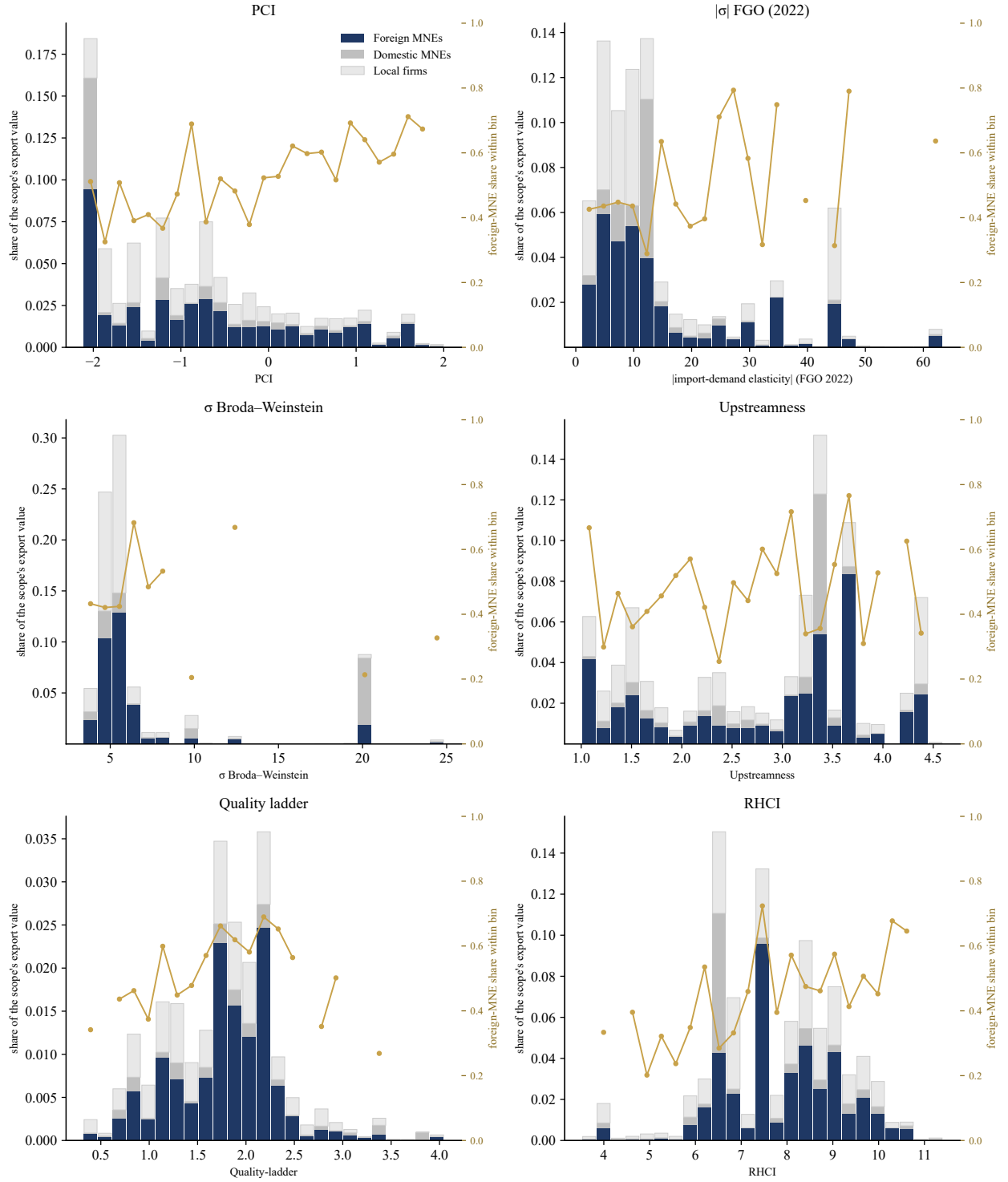
### 3.3 Variant: other measures of product sophistication; substitution elasticities (item 1b)



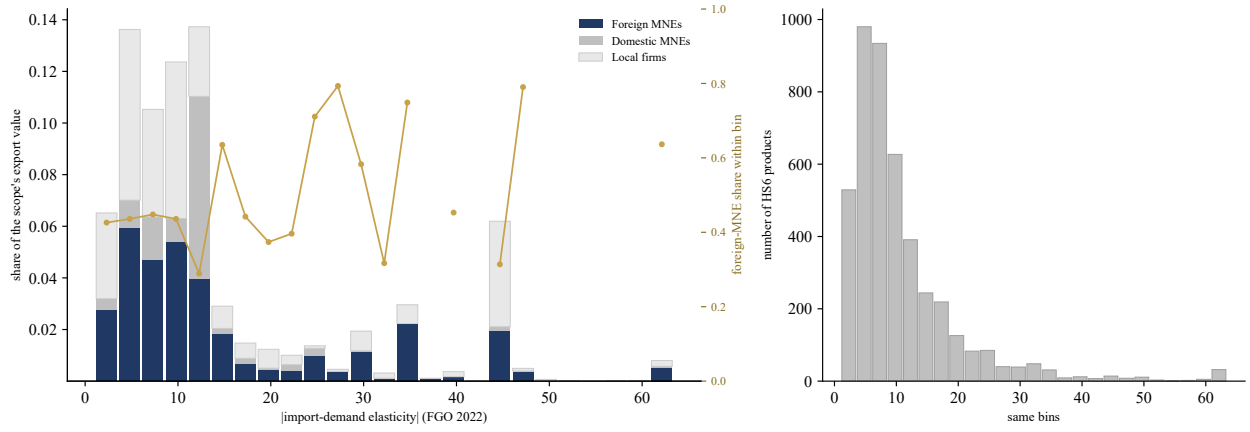
**Figure 9:** Figure 2 redrawn for six sophistication measures (foreign navy, domestic gray; quintiles over the scope's HS6 products).



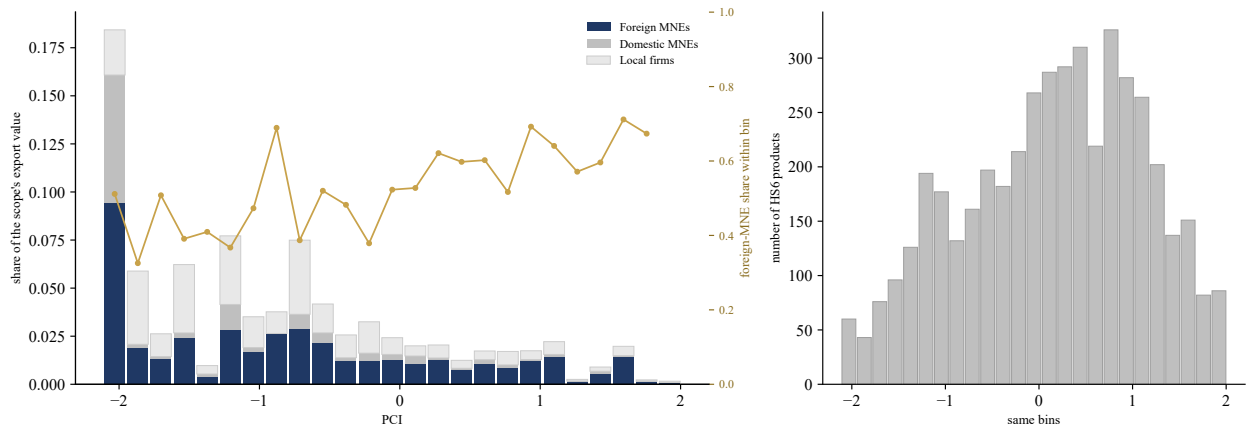
**Figure 10:** Figure 2 with  $|\sigma|$  from Fontagné, Guimbard and Orefice (2022) in place of the PCI.



**Figure 11:** Full distributions: export value across bins of each measure, stacked by owner type, with the foreign-MNE share of each bin (line, right axis).



**Figure 12:** Distribution over  $|\sigma|$  (FGO 2022): export value by owner type and foreign share per bin (left); number of HS6 products per bin (right).



**Figure 13:** Distribution over the PCI: export value by owner type and foreign share per bin (left); number of HS6 products per bin (right).

**Table 2:** Foreign and domestic MNE shares by quintile, six sophistication measures

| Measure                  | Foreign MNE share |      |      |      |      | Domestic MNE share |      |      |      |      |
|--------------------------|-------------------|------|------|------|------|--------------------|------|------|------|------|
|                          | Q1                | Q2   | Q3   | Q4   | Q5   | Q1                 | Q2   | Q3   | Q4   | Q5   |
| PCI                      | 0.46              | 0.44 | 0.56 | 0.60 | 0.65 | 0.18               | 0.10 | 0.13 | 0.09 | 0.07 |
| $ \sigma $ FGO (2022)    | 0.46              | 0.38 | 0.45 | 0.34 | 0.52 | 0.06               | 0.15 | 0.07 | 0.40 | 0.06 |
| $\sigma$ Broda–Weinstein | 0.44              | 0.40 | 0.40 | 0.48 | 0.39 | 0.15               | 0.10 | 0.06 | 0.06 | 0.38 |
| Upstreamness             | 0.52              | 0.41 | 0.41 | 0.47 | 0.49 | 0.06               | 0.10 | 0.16 | 0.09 | 0.21 |
| Quality ladder           | 0.49              | 0.49 | 0.65 | 0.64 | 0.43 | 0.08               | 0.08 | 0.07 | 0.07 | 0.16 |
| RHCI                     | 0.34              | 0.59 | 0.51 | 0.49 | 0.53 | 0.25               | 0.06 | 0.06 | 0.12 | 0.11 |

Value-weighted shares of export value within quintile; quintiles over HS6 products (unweighted); ‘–’ = fewer than 30 HS6 products carry the measure in this scope.  $|\sigma|$  FGO: absolute value of the tariff-based import-demand elasticity of Fontagné, Guimbard and Orefice (2022), HS6 rev. 2007 (non-significant or positive HS6 estimates replaced by the HS4 average by the source); larger = more substitutable.  $\sigma$  Broda–Weinstein, upstreamness (Antràs–Chor), quality ladder (Khandelwal), RHCI (UNCTAD) as in the document’s appendix.

**Table 3:** Correlations across sophistication measures (HS6, value-weighted)

|                          | PCI   | $ \sigma $ FGO (2022) | $\sigma$ Broda–Weinstein | Upstreamness | Quality ladder | RHCI  |
|--------------------------|-------|-----------------------|--------------------------|--------------|----------------|-------|
| PCI                      | 1.00  | -0.22                 | -0.49                    | -0.49        | 0.28           | 0.74  |
| $ \sigma $ FGO (2022)    | -0.22 | 1.00                  | 0.03                     | 0.62         | 0.12           | 0.02  |
| $\sigma$ Broda–Weinstein | -0.49 | 0.03                  | 1.00                     | 0.20         | -0.11          | -0.25 |
| Upstreamness             | -0.49 | 0.62                  | 0.20                     | 1.00         | 0.12           | -0.19 |
| Quality ladder           | 0.28  | 0.12                  | -0.11                    | 0.12         | 1.00           | 0.34  |
| RHCI                     | 0.74  | 0.02                  | -0.25                    | -0.19        | 0.34           | 1.00  |

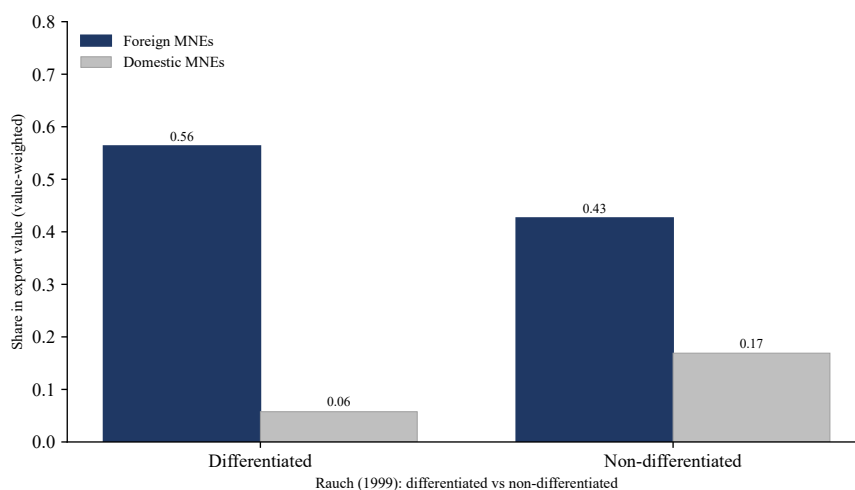
Value-weighted Pearson correlations, pairwise: each cell uses the HS6 products carrying both measures (3,124 to 4,841 products; ‘–’ = fewer than 30).

**Table 4:** MNE shares by Rauch (1999) class

| Rauch (1999) class     | Foreign | Domestic | MNE total | N HS6 | Value (\$bn) |
|------------------------|---------|----------|-----------|-------|--------------|
| Differentiated         | 0.563   | 0.058    | 0.621     | 2,712 | 450.4        |
| Reference-priced       | 0.549   | 0.082    | 0.631     | 1,636 | 698.5        |
| Homogeneous (exchange) | 0.343   | 0.228    | 0.571     | 319   | 1,030.0      |

**Table 5:** MNE shares by Rauch (1999) class, two classes: differentiated vs non-differentiated (reference-priced + homogeneous)

| Rauch (1999): differentiated vs non-differentiated | Foreign | Domestic | MNE total | N HS6 | Value (\$bn) |
|----------------------------------------------------|---------|----------|-----------|-------|--------------|
| Differentiated                                     | 0.563   | 0.058    | 0.621     | 2,712 | 450.4        |
| Non-differentiated                                 | 0.426   | 0.169    | 0.595     | 1,955 | 1,728.4      |

**Figure 14:** Foreign vs domestic MNE shares of export value: differentiated vs non-differentiated products (Rauch 1999; non-differentiated = reference-priced + homogeneous).**Table 6:** MNE shares by BEC end use

| BEC end use            | Foreign | Domestic | MNE total | N HS6 | Value (\$bn) |
|------------------------|---------|----------|-----------|-------|--------------|
| Intermediate           | 0.476   | 0.155    | 0.631     | 3,072 | 1,667.8      |
| Consumption            | 0.409   | 0.087    | 0.496     | 1,193 | 449.1        |
| Capital                | 0.647   | 0.056    | 0.703     | 659   | 88.5         |
| Mixed (passenger cars) | 0.885   | 0.000    | 0.885     | 8     | 28.9         |
| n.e.s.                 | 0.198   | 0.345    | 0.543     | 20    | 62.6         |

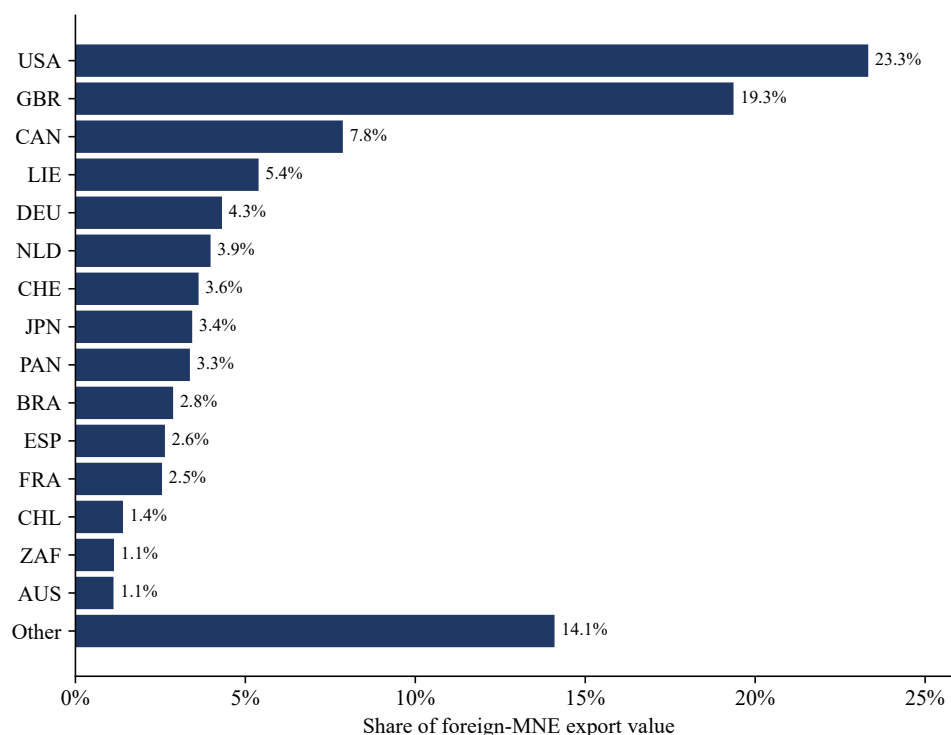
**Table 7:** Note's Table A.4 ladder with the FGO elasticity added (ODPY cells, MNE value share)

|                                           | (1)                    | (2)                    | (3)                    |
|-------------------------------------------|------------------------|------------------------|------------------------|
| <i>Panel A: MNE<sub>total</sub> share</i> |                        |                        |                        |
| Complexity (PCI)                          | 0.0281***<br>(0.0047)  | 0.0307***<br>(0.0045)  | 0.0409***<br>(0.0040)  |
| $\sigma$   FGO (per s.d.)                 | -0.0078***<br>(0.0026) | -0.0074***<br>(0.0025) | -0.0057***<br>(0.0021) |
| Upstreamness                              | 0.0087*<br>(0.0047)    | 0.0079*<br>(0.0044)    | 0.0108***<br>(0.0038)  |
| <i>Panel B: MNE<sub>ext</sub> share</i>   |                        |                        |                        |
| Complexity (PCI)                          | 0.0926***<br>(0.0049)  | 0.0925***<br>(0.0047)  | 0.0855***<br>(0.0043)  |
| $\sigma$   FGO (per s.d.)                 | 0.0339***<br>(0.0023)  | 0.0331***<br>(0.0022)  | 0.0241***<br>(0.0020)  |
| Upstreamness                              | -0.0239***<br>(0.0043) | -0.0253***<br>(0.0041) | -0.0148***<br>(0.0034) |
| <i>Panel C: MNE<sub>dom</sub> share</i>   |                        |                        |                        |
| Complexity (PCI)                          | -0.0645***<br>(0.0053) | -0.0618***<br>(0.0048) | -0.0446***<br>(0.0038) |
| $\sigma$   FGO (per s.d.)                 | -0.0417***<br>(0.0027) | -0.0405***<br>(0.0025) | -0.0298***<br>(0.0022) |
| Upstreamness                              | 0.0326***<br>(0.0041)  | 0.0332***<br>(0.0037)  | 0.0256***<br>(0.0028)  |
| Origin FE                                 | ✓                      |                        |                        |
| Destination FE                            | ✓                      |                        |                        |
| Year FE                                   | ✓                      |                        |                        |
| Origin $\times$ year FE                   |                        | ✓                      |                        |
| Destination $\times$ year FE              |                        | ✓                      |                        |
| Origin $\times$ dest. $\times$ year FE    |                        |                        | ✓                      |
| Observations                              | 1,563,005              | 1,562,745              | 1,561,136              |

Origin-destination-product-year cells; dep. var. MNE value share; weighted by total trade value; robust (HC1) SE. | $\sigma$ | FGO standardised (mean 0, s.d. 1). \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

## 4 Fact 3 — Multinational corporations from a small set of countries dominate exports

### 4.1 Original exhibit



**Figure 15:** Figure 4 of the note, reproduced on the current base: foreign-MNE export value by parent country, top 15 plus Other.

*Note:* Foreign MNEs with a recorded parent country.

**Table 8:** Foreign-MNE export value by parent country (numbers behind the figure)

| Parent country                                | Share (%) | Value (\$bn) |
|-----------------------------------------------|-----------|--------------|
| USA                                           | 23.3      | 238.9        |
| GBR                                           | 19.3      | 198.3        |
| CAN                                           | 7.8       | 80.4         |
| LIE                                           | 5.4       | 55.0         |
| DEU                                           | 4.3       | 43.9         |
| NLD                                           | 3.9       | 40.5         |
| CHE                                           | 3.6       | 36.9         |
| JPN                                           | 3.4       | 34.9         |
| PAN                                           | 3.3       | 34.2         |
| BRA                                           | 2.8       | 29.2         |
| ESP                                           | 2.6       | 26.7         |
| FRA                                           | 2.5       | 25.8         |
| CHL                                           | 1.4       | 14.1         |
| ZAF                                           | 1.1       | 11.3         |
| AUS                                           | 1.1       | 11.2         |
| Other (150 countries)                         | 14.1      | 144.3        |
| Foreign MNEs with recorded parent             | 100.0     | 1,025.6      |
| Foreign MNEs, parent unknown (excluded above) | —         | 83.6         |

## 4.2 Variant: parent country × export destination — tables (item 1c)

**Table 9:** Parent region × destination region, export value (USD bn)

| Parent region / Destination region | Latin America | North America | Europe | Asia  | Africa | Oceania | Rest of World | Total   |
|------------------------------------|---------------|---------------|--------|-------|--------|---------|---------------|---------|
| Latin America                      | 49.5          | 19.0          | 18.1   | 33.4  | 1.5    | 0.9     | 0.4           | 122.6   |
| North America                      | 100.0         | 75.5          | 63.9   | 73.7  | 5.6    | 0.3     | 0.3           | 319.3   |
| Europe                             | 136.8         | 57.3          | 87.2   | 188.2 | 14.3   | 5.1     | 0.6           | 489.5   |
| Asia                               | 35.1          | 12.9          | 8.1    | 11.7  | 0.8    | 0.1     | 0.2           | 68.8    |
| Africa                             | 1.5           | 0.9           | 3.7    | 7.4   | 0.3    | 0.0     | 0.0           | 13.9    |
| Oceania                            | 2.3           | 0.6           | 0.9    | 7.3   | 0.0    | 0.0     | 0.0           | 11.2    |
| Rest of World                      | 0.1           | 0.1           | 0.0    | 0.0   | 0.0    | 0.0     | 0.0           | 0.2     |
| Foreign MNE, parent unknown        | 32.7          | 18.7          | 15.6   | 14.3  | 1.1    | 0.4     | 0.8           | 83.6    |
| Domestic MNEs                      | 87.6          | 111.4         | 37.7   | 91.5  | 7.7    | 1.7     | 0.9           | 338.6   |
| Local firms (unmatched)            | 256.8         | 204.0         | 205.3  | 243.5 | 27.0   | 5.7     | 3.6           | 946.0   |
| Total                              | 702.4         | 500.3         | 440.6  | 671.1 | 58.2   | 14.2    | 6.9           | 2,393.8 |

Export value, USD bn. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

**Table 10:** Parent region × destination region, row %

| Parent region / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|------------------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Latin America                      | 40.4          | 15.5          | 14.7   | 27.2 | 1.2    | 0.7     | 0.3           | 100.0 |
| North America                      | 31.3          | 23.6          | 20.0   | 23.1 | 1.7    | 0.1     | 0.1           | 100.0 |
| Europe                             | 28.0          | 11.7          | 17.8   | 38.4 | 2.9    | 1.0     | 0.1           | 100.0 |
| Asia                               | 51.0          | 18.7          | 11.7   | 17.0 | 1.1    | 0.2     | 0.3           | 100.0 |
| Africa                             | 10.9          | 6.6           | 26.9   | 53.2 | 2.2    | 0.0     | 0.2           | 100.0 |
| Oceania                            | 20.1          | 5.7           | 8.4    | 65.5 | 0.1    | 0.2     | 0.0           | 100.0 |
| Rest of World                      | 32.6          | 25.0          | 20.9   | 14.3 | 0.0    | 7.1     | 0.0           | 100.0 |
| Foreign MNE, parent unknown        | 39.1          | 22.4          | 18.7   | 17.1 | 1.3    | 0.4     | 1.0           | 100.0 |
| Domestic MNEs                      | 25.9          | 32.9          | 11.1   | 27.0 | 2.3    | 0.5     | 0.3           | 100.0 |
| Local firms (unmatched)            | 27.1          | 21.6          | 21.7   | 25.7 | 2.9    | 0.6     | 0.4           | 100.0 |

Row percentages: destination mix of each row group. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

**Table 11:** Parent region × destination region, column %

| Parent region / Destination region | Latin America | North America | Europe | Asia  | Africa | Oceania | Rest of World |
|------------------------------------|---------------|---------------|--------|-------|--------|---------|---------------|
| Latin America                      | 7.0           | 3.8           | 4.1    | 5.0   | 2.5    | 6.1     | 5.5           |
| North America                      | 14.2          | 15.1          | 14.5   | 11.0  | 9.6    | 1.8     | 4.3           |
| Europe                             | 19.5          | 11.4          | 19.8   | 28.0  | 24.5   | 36.1    | 9.1           |
| Asia                               | 5.0           | 2.6           | 1.8    | 1.7   | 1.3    | 0.9     | 2.8           |
| Africa                             | 0.2           | 0.2           | 0.9    | 1.1   | 0.5    | 0.0     | 0.4           |
| Oceania                            | 0.3           | 0.1           | 0.2    | 1.1   | 0.0    | 0.1     | 0.0           |
| Rest of World                      | 0.0           | 0.0           | 0.0    | 0.0   | 0.0    | 0.1     | 0.0           |
| Foreign MNE, parent unknown        | 4.7           | 3.7           | 3.5    | 2.1   | 1.9    | 2.6     | 12.3          |
| Domestic MNEs                      | 12.5          | 22.3          | 8.6    | 13.6  | 13.3   | 11.9    | 13.1          |
| Local firms (unmatched)            | 36.6          | 40.8          | 46.6   | 36.3  | 46.4   | 40.4    | 52.5          |
| Total                              | 100.0         | 100.0         | 100.0  | 100.0 | 100.0  | 100.0   | 100.0         |

Column percentages: who supplies each destination. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B). Rows for domestic MNEs and local exporters are given for comparison.

**Table 12:** Top-10 parents × top-10 destinations, export value (USD bn)

| Parent / Destination | CHN   | USA   | BRA   | JPN  | ESP  | CHL  | KOR  | CAN  | NLD  | ARG  | Other destinations | Total   |
|----------------------|-------|-------|-------|------|------|------|------|------|------|------|--------------------|---------|
| USA                  | 24.1  | 53.4  | 18.6  | 9.0  | 9.5  | 6.9  | 5.0  | 1.2  | 8.2  | 11.7 | 91.4               | 238.9   |
| GBR                  | 72.4  | 11.2  | 7.7   | 14.0 | 5.9  | 6.7  | 12.8 | 6.5  | 6.4  | 0.4  | 54.2               | 198.3   |
| CAN                  | 10.3  | 5.3   | 3.4   | 4.3  | 2.5  | 0.4  | 1.5  | 15.7 | 0.7  | 0.2  | 36.2               | 80.4    |
| LIE                  | 13.7  | 2.0   | 2.0   | 10.3 | 1.9  | 1.4  | 3.0  | 0.0  | 1.0  | 0.0  | 19.6               | 55.0    |
| DEU                  | 0.4   | 6.5   | 14.0  | 0.5  | 0.8  | 1.4  | 0.3  | 0.5  | 0.4  | 0.4  | 18.8               | 43.9    |
| NLD                  | 3.7   | 1.6   | 10.9  | 0.2  | 0.4  | 1.5  | 0.5  | 0.1  | 0.9  | 1.3  | 19.6               | 40.5    |
| CHE                  | 2.3   | 1.8   | 2.5   | 0.2  | 1.1  | 1.3  | 0.3  | 0.2  | 0.7  | 0.2  | 26.3               | 36.9    |
| JPN                  | 0.8   | 3.0   | 15.7  | 3.1  | 0.2  | 0.5  | 0.1  | 0.2  | 0.3  | 0.7  | 10.3               | 34.9    |
| PAN                  | 12.0  | 3.3   | 0.4   | 0.6  | 0.6  | 0.4  | 2.1  | 0.2  | 0.7  | 0.1  | 13.9               | 34.2    |
| BRA                  | 3.8   | 4.1   | 5.4   | 0.5  | 0.2  | 0.9  | 0.2  | 0.2  | 0.6  | 0.7  | 12.7               | 29.2    |
| Other parents        | 22.2  | 45.6  | 27.3  | 5.2  | 8.6  | 9.3  | 3.7  | 3.8  | 4.9  | 8.7  | 93.9               | 233.4   |
| Total                | 165.8 | 137.8 | 107.9 | 47.9 | 31.6 | 30.7 | 29.5 | 28.5 | 25.0 | 24.3 | 396.9              | 1,025.6 |

Export value, USD bn. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

**Table 13:** Top-10 parents × top-10 destinations, row %

| Parent / Destination | CHN  | USA  | BRA  | JPN  | ESP | CHL | KOR | CAN  | NLD | ARG | Other destinations | Total |
|----------------------|------|------|------|------|-----|-----|-----|------|-----|-----|--------------------|-------|
| USA                  | 10.1 | 22.3 | 7.8  | 3.8  | 4.0 | 2.9 | 2.1 | 0.5  | 3.5 | 4.9 | 38.2               | 100.0 |
| GBR                  | 36.5 | 5.6  | 3.9  | 7.1  | 3.0 | 3.4 | 6.5 | 3.3  | 3.2 | 0.2 | 27.3               | 100.0 |
| CAN                  | 12.8 | 6.6  | 4.2  | 5.3  | 3.1 | 0.5 | 1.9 | 19.5 | 0.9 | 0.2 | 45.0               | 100.0 |
| LIE                  | 24.9 | 3.7  | 3.7  | 18.7 | 3.5 | 2.5 | 5.4 | 0.0  | 1.8 | 0.0 | 35.6               | 100.0 |
| DEU                  | 1.0  | 14.8 | 31.9 | 1.1  | 1.7 | 3.1 | 0.6 | 1.1  | 1.0 | 0.9 | 42.8               | 100.0 |
| NLD                  | 9.2  | 3.8  | 26.9 | 0.4  | 1.0 | 3.6 | 1.1 | 0.2  | 2.2 | 3.1 | 48.4               | 100.0 |
| CHE                  | 6.2  | 5.0  | 6.8  | 0.5  | 2.9 | 3.5 | 0.8 | 0.4  | 2.0 | 0.5 | 71.4               | 100.0 |
| JPN                  | 2.4  | 8.7  | 45.0 | 8.9  | 0.4 | 1.4 | 0.4 | 0.5  | 0.9 | 1.9 | 29.5               | 100.0 |
| PAN                  | 35.0 | 9.5  | 1.2  | 1.8  | 1.7 | 1.1 | 6.2 | 0.6  | 2.0 | 0.2 | 40.7               | 100.0 |
| BRA                  | 12.9 | 14.1 | 18.6 | 1.6  | 0.6 | 3.1 | 0.6 | 0.6  | 2.1 | 2.3 | 43.4               | 100.0 |
| Other parents        | 9.5  | 19.5 | 11.7 | 2.2  | 3.7 | 4.0 | 1.6 | 1.6  | 2.1 | 3.7 | 40.3               | 100.0 |

Row percentages: destination mix of each row group. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

**Table 14:** Top-10 parents × top-10 destinations, column %

| Parent / Destination | CHN   | USA   | BRA   | JPN   | ESP   | CHL   | KOR   | CAN   | NLD   | ARG   | Other destinations |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------------------|
| USA                  | 14.6  | 38.7  | 17.2  | 18.8  | 30.1  | 22.5  | 17.0  | 4.1   | 33.0  | 48.1  | 23.0               |
| GBR                  | 43.7  | 8.1   | 7.2   | 29.3  | 18.8  | 21.8  | 43.5  | 22.8  | 25.7  | 1.5   | 13.7               |
| CAN                  | 6.2   | 3.8   | 3.1   | 9.0   | 7.8   | 1.4   | 5.1   | 55.1  | 3.0   | 0.8   | 9.1                |
| LIE                  | 8.3   | 1.5   | 1.9   | 21.5  | 6.1   | 4.6   | 10.1  | 0.1   | 4.0   | 0.1   | 4.9                |
| DEU                  | 0.3   | 4.7   | 13.0  | 1.0   | 2.4   | 4.5   | 0.9   | 1.7   | 1.8   | 1.6   | 4.7                |
| NLD                  | 2.3   | 1.1   | 10.1  | 0.3   | 1.3   | 4.7   | 1.6   | 0.2   | 3.5   | 5.2   | 4.9                |
| CHE                  | 1.4   | 1.3   | 2.3   | 0.4   | 3.3   | 4.3   | 1.0   | 0.6   | 2.9   | 0.8   | 6.6                |
| JPN                  | 0.5   | 2.2   | 14.6  | 6.5   | 0.5   | 1.6   | 0.4   | 0.6   | 1.2   | 2.7   | 2.6                |
| PAN                  | 7.2   | 2.4   | 0.4   | 1.3   | 1.8   | 1.2   | 7.2   | 0.8   | 2.8   | 0.2   | 3.5                |
| BRA                  | 2.3   | 3.0   | 5.0   | 1.0   | 0.6   | 2.9   | 0.6   | 0.6   | 2.5   | 2.8   | 3.2                |
| Other parents        | 13.4  | 33.1  | 25.3  | 10.9  | 27.3  | 30.4  | 12.6  | 13.4  | 19.6  | 36.0  | 23.7               |
| Total                | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0 | 100.0              |

Column percentages: who supplies each destination. Foreign MNEs with a recorded parent country; top 10 parents and top 10 destinations by foreign-MNE export value. Pooled 2006–2022, nine LAC origins (Ecuador excluded), all destinations. Parent = country of the multinational's ultimate parent as recorded (Orbis/D&B).

**Table 15:** Foreign-MNE exports: destination region by LAC origin (row %)

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 45.5          | 9.0           | 15.5   | 23.0 | 6.2    | 0.8     | 0.0           | 292.3 |
| CHL    |                    | 13.1          | 11.6          | 13.3   | 60.5 | 0.3    | 1.2     | 0.0           | 229.3 |
| COL    |                    | 40.7          | 19.7          | 25.4   | 12.6 | 0.5    | 0.3     | 0.9           | 186.5 |
| CRI    |                    | 33.0          | 40.7          | 16.2   | 9.8  | 0.1    | 0.1     | 0.0           | 62.5  |
| DOM    |                    | 20.3          | 59.3          | 9.4    | 10.5 | 0.2    | 0.2     | 0.1           | 47.1  |
| PER    |                    | 15.4          | 12.9          | 25.3   | 44.9 | 1.0    | 0.4     | 0.0           | 183.3 |
| PRY    |                    | 79.4          | 1.3           | 11.6   | 6.6  | 1.1    | 0.0     | 0.0           | 40.0  |
| SLV    |                    | 46.8          | 48.1          | 4.3    | 0.6  | 0.1    | 0.1     | 0.0           | 31.0  |
| URY    |                    | 39.1          | 7.7           | 20.5   | 28.0 | 3.0    | 0.1     | 1.6           | 37.2  |

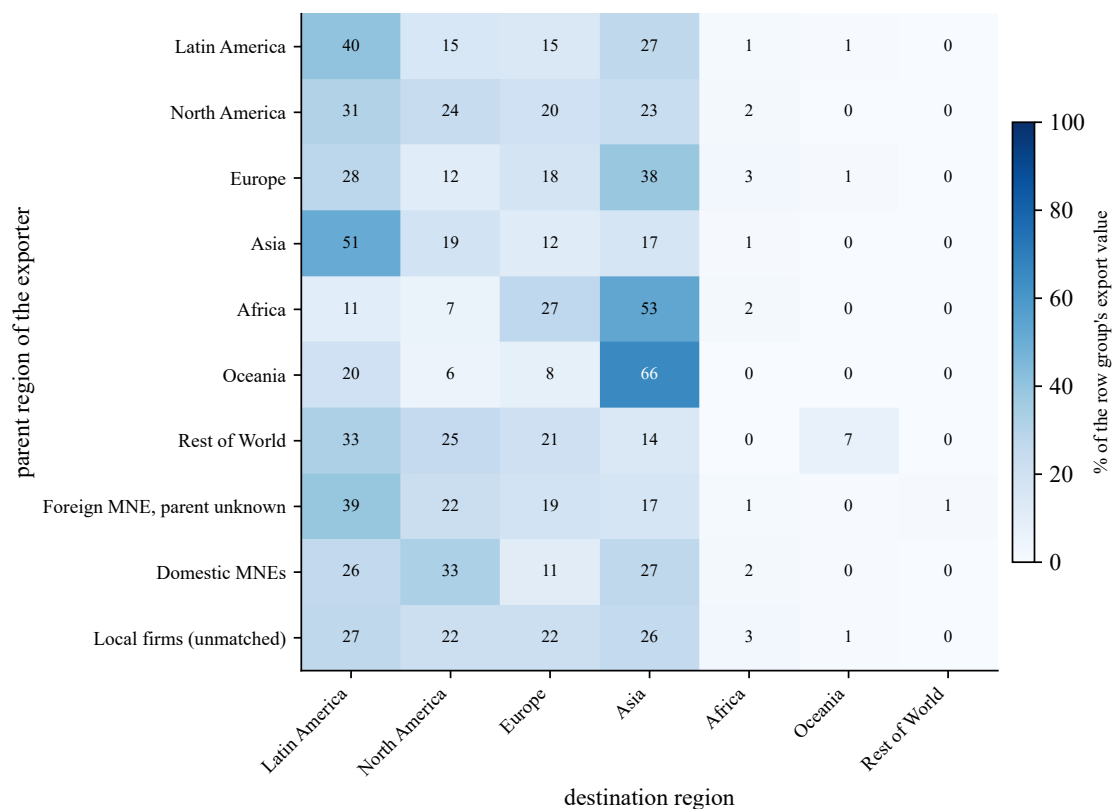
Foreign-MNE exports; row percentages; last column = row total, USD bn.

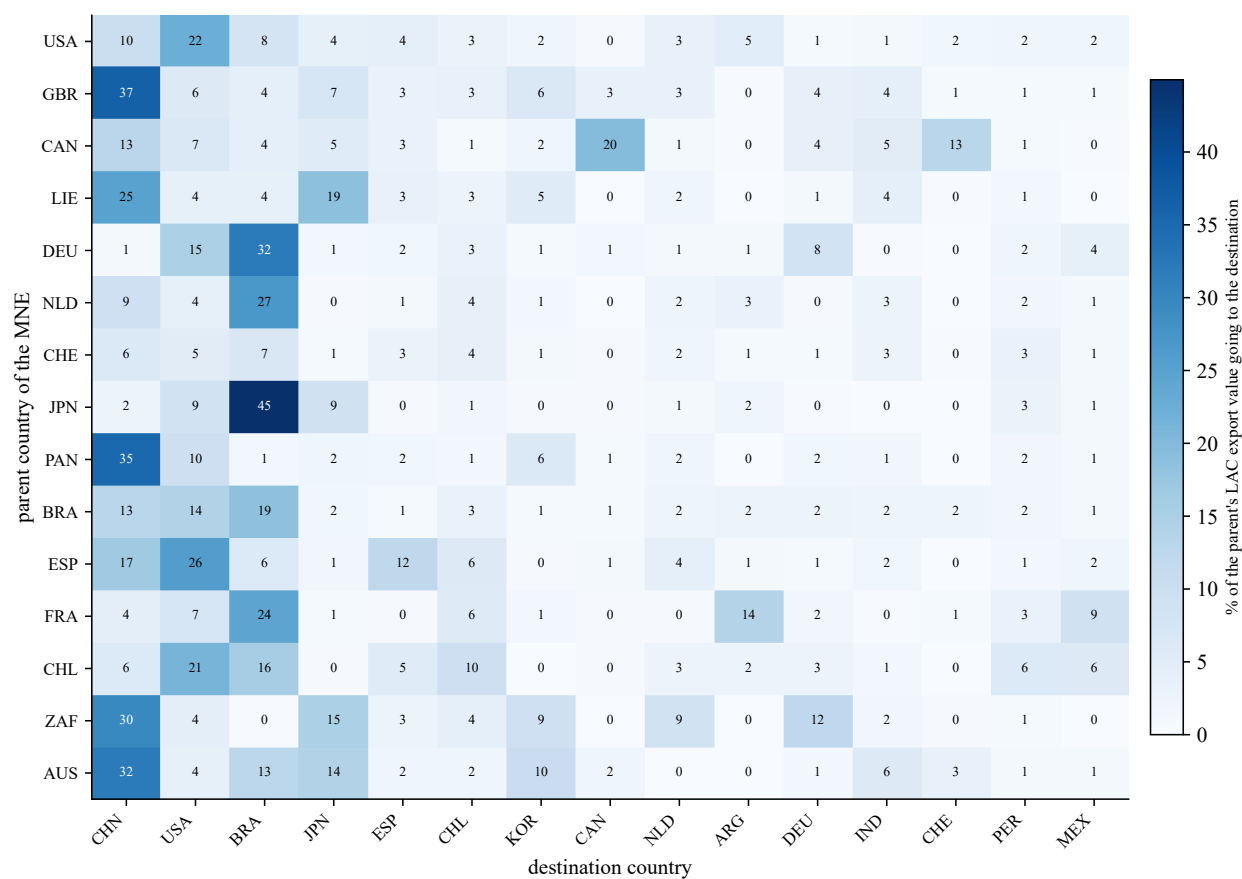
**Table 16:** Local (unmatched) firms: destination region by LAC origin (row %)

| Origin | Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|--------|--------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| ARG    |                    | 32.2          | 8.3           | 21.4   | 28.5 | 8.4    | 1.3     | 0.0           | 260.8 |
| CHL    |                    | 15.5          | 18.8          | 20.4   | 44.3 | 0.6    | 0.5     | 0.0           | 184.3 |
| COL    |                    | 32.2          | 32.8          | 23.5   | 8.5  | 0.8    | 0.3     | 1.9           | 135.0 |
| CRI    |                    | 32.1          | 35.4          | 26.3   | 5.4  | 0.3    | 0.5     | 0.0           | 38.4  |
| DOM    |                    | 31.9          | 46.8          | 15.7   | 4.6  | 0.6    | 0.0     | 0.3           | 20.9  |
| PER    |                    | 18.3          | 28.0          | 24.6   | 28.1 | 0.3    | 0.4     | 0.2           | 221.5 |
| PRY    |                    | 64.6          | 3.6           | 20.0   | 10.6 | 1.1    | 0.0     | 0.0           | 24.7  |
| SLV    |                    | 44.9          | 46.3          | 4.6    | 3.7  | 0.1    | 0.3     | 0.1           | 34.2  |
| URY    |                    | 38.7          | 5.2           | 21.6   | 26.0 | 7.3    | 0.1     | 1.3           | 26.2  |

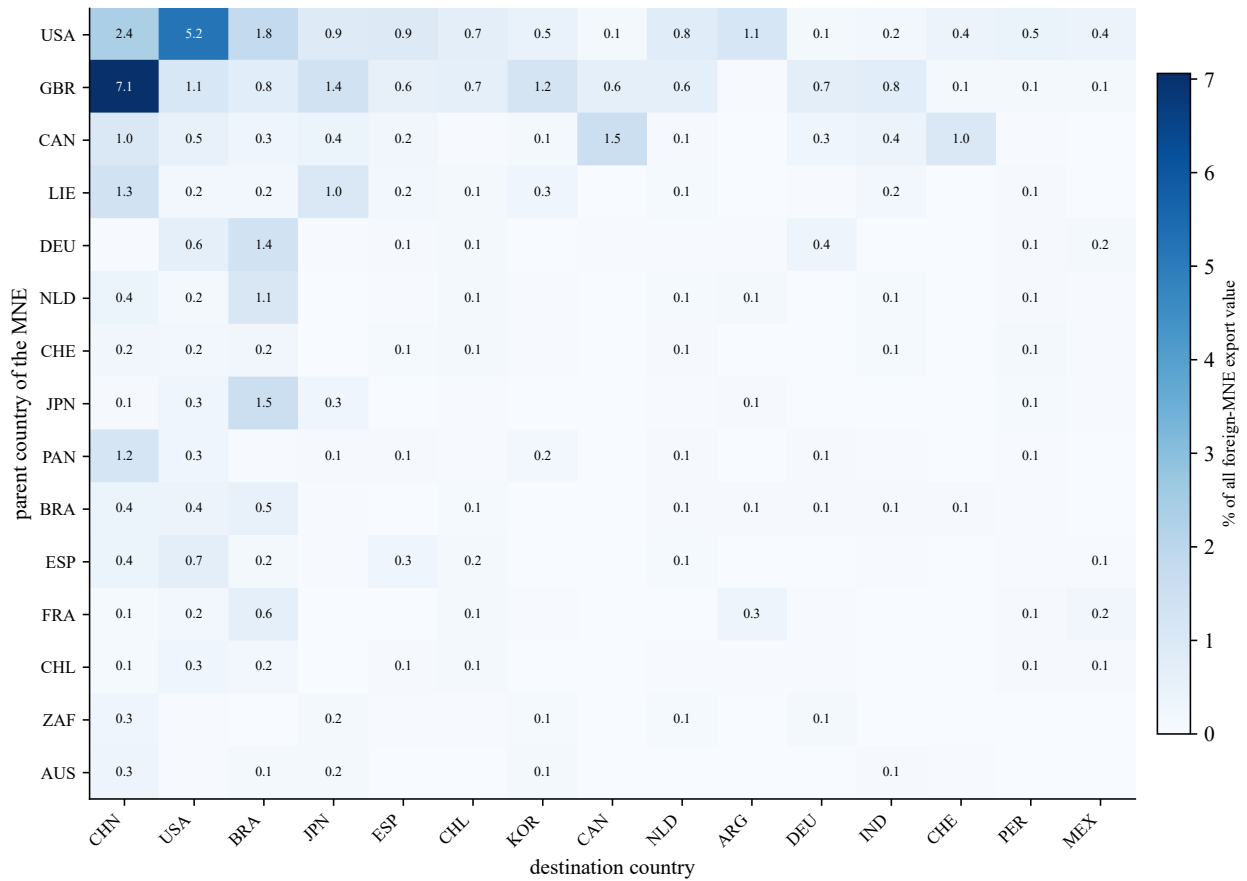
Local (unmatched) firms exports; row percentages; last column = row total, USD bn.

### 4.3 Variant: heat maps (item 1d)

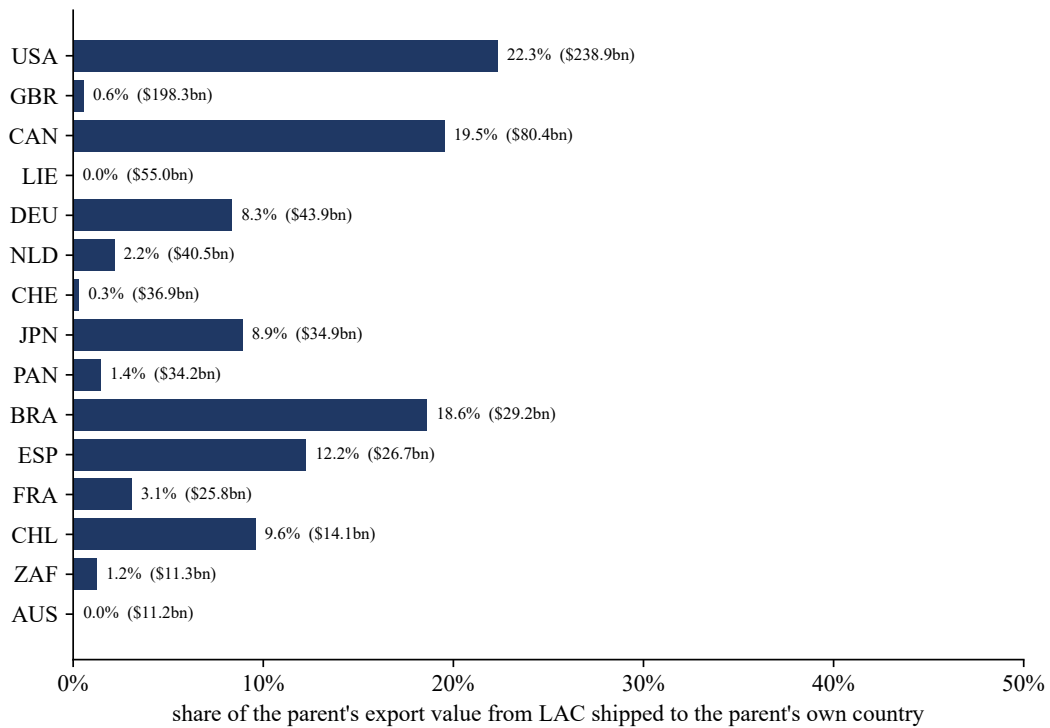
**Figure 16:** Parent region × destination region, row %.



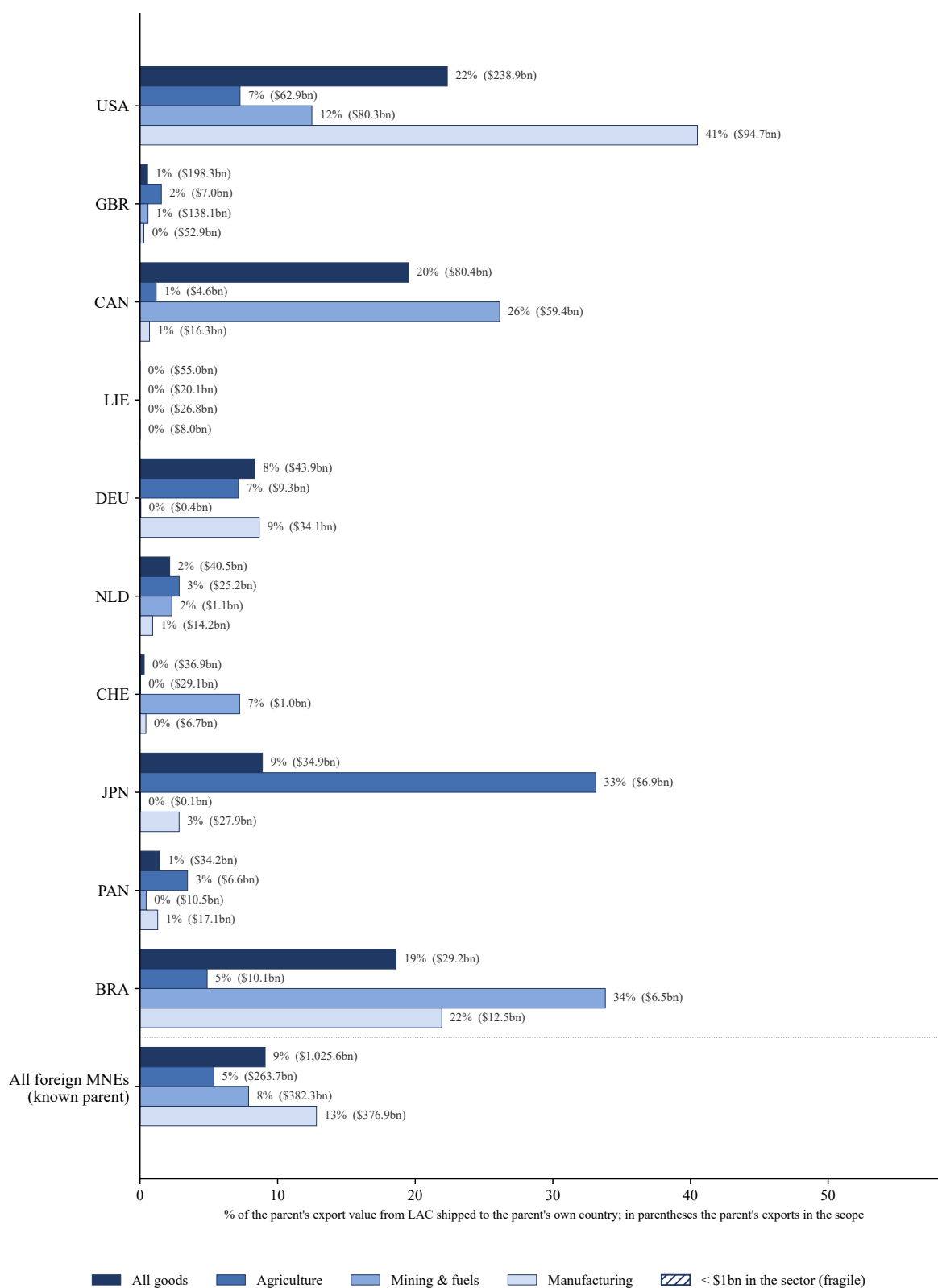
**Figure 17:** Top-15 parents × top-15 destinations: % of the parent's export value going to each destination.



**Figure 18:** Top-15 parents × top-15 destinations: cell share of all foreign-MNE export value (%).



**Figure 19:** Share of each parent's export value shipped to the parent's own country.



**Figure 20:** Share of each parent's export value shipped to the parent's own country: all goods and the three sectors, top-10 parents.

*Note:* Hatched bars: the parent exports less than \$1bn in the sector, so the share is fragile; the parent's exports in each scope are in parentheses.

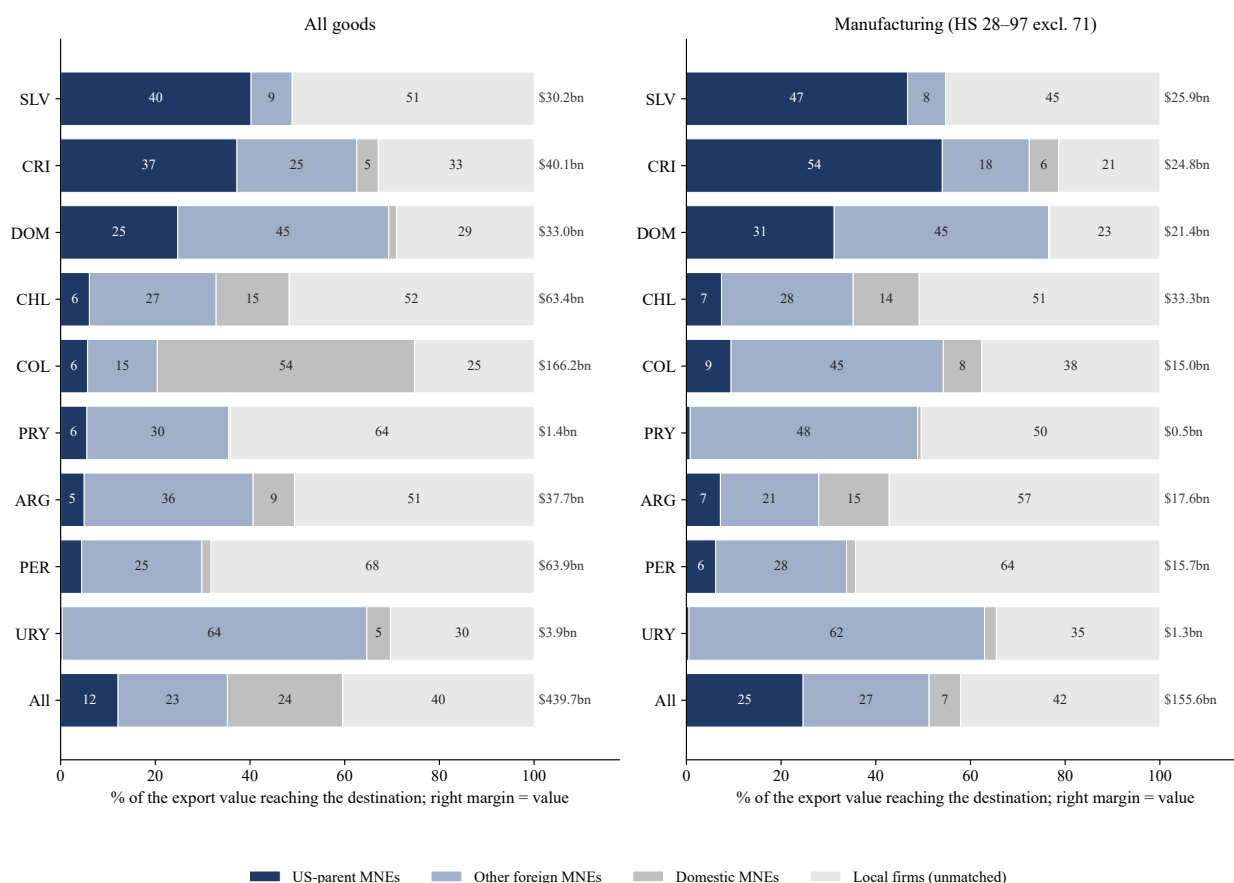
**Table 17: Home share by parent, all goods and by sector (numbers behind the figure)**

| Parent                        | % of the parent's LAC exports shipped to the parent's own country |             |                |               | Parent's LAC exports (\$bn) |             |                |               |
|-------------------------------|-------------------------------------------------------------------|-------------|----------------|---------------|-----------------------------|-------------|----------------|---------------|
|                               | All goods                                                         | Agriculture | Mining & fuels | Manufacturing | All goods                   | Agriculture | Mining & fuels | Manufacturing |
| USA                           | 22.3                                                              | 7.3         | 12.5           | 40.5          | 238.9                       | 62.9        | 80.3           | 94.7          |
| GBR                           | 0.6                                                               | 1.6         | 0.6            | 0.3           | 198.3                       | 7.0         | 138.1          | 52.9          |
| CAN                           | 19.5                                                              | 1.2         | 26.1           | 0.7           | 80.4                        | 4.6         | 59.4           | 16.3          |
| LIE                           | 0.0                                                               | 0.0         | 0.0            | 0.0           | 55.0                        | 20.1        | 26.8           | 8.0           |
| DEU                           | 8.3                                                               | 7.1         | 0.1            | 8.7           | 43.9                        | 9.3         | 0.4            | 34.1          |
| NLD                           | 2.2                                                               | 2.8         | 2.3            | 0.9           | 40.5                        | 25.2        | 1.1            | 14.2          |
| CHE                           | 0.3                                                               | 0.0         | 7.2            | 0.4           | 36.9                        | 29.1        | 1.0            | 6.7           |
| JPN                           | 8.9                                                               | 33.1        | 0.0            | 2.8           | 34.9                        | 6.9         | 0.1            | 27.9          |
| PAN                           | 1.4                                                               | 3.5         | 0.4            | 1.3           | 34.2                        | 6.6         | 10.5           | 17.1          |
| BRA                           | 18.6                                                              | 4.9         | 33.8           | 21.9          | 29.2                        | 10.1        | 6.5            | 12.5          |
| All known-parent foreign MNEs | 9.1                                                               | 5.4         | 7.9            | 12.8          | 1,025.6                     | 263.7       | 382.3          | 376.9         |

Top-10 parent countries by foreign-MNE export value (all goods). Home share = value the parent's affiliates export from the nine origins to the parent's country, over everything they export from the nine origins, all goods and within each sector. Shares computed on less than \$1bn of exports are fragile (see the value columns). Pooled 2006–2022; foreign MNEs with a recorded parent country.

#### 4.4 Variant: seen from the destination — who carries the exports that reach each market (revision 6)

The exhibits above ask where each parent ships. These turn the question around: of the value that reaches a destination, how much is moved by multinationals whose parent sits in that destination, by US-parent MNEs, by other foreign MNEs, by domestic MNEs and by local (unmatched) firms.



**Figure 21: Exports to the United States by origin: share moved by US-parent MNEs, other foreign MNEs, domestic MNEs and local firms — all goods (left) and manufacturing (right).**

*Note:* Origins sorted by the all-goods US-parent share; right margin = the origin's exports to the USA in the scope.

**Table 18:** Exports to the United States by sector: who carries them (%)

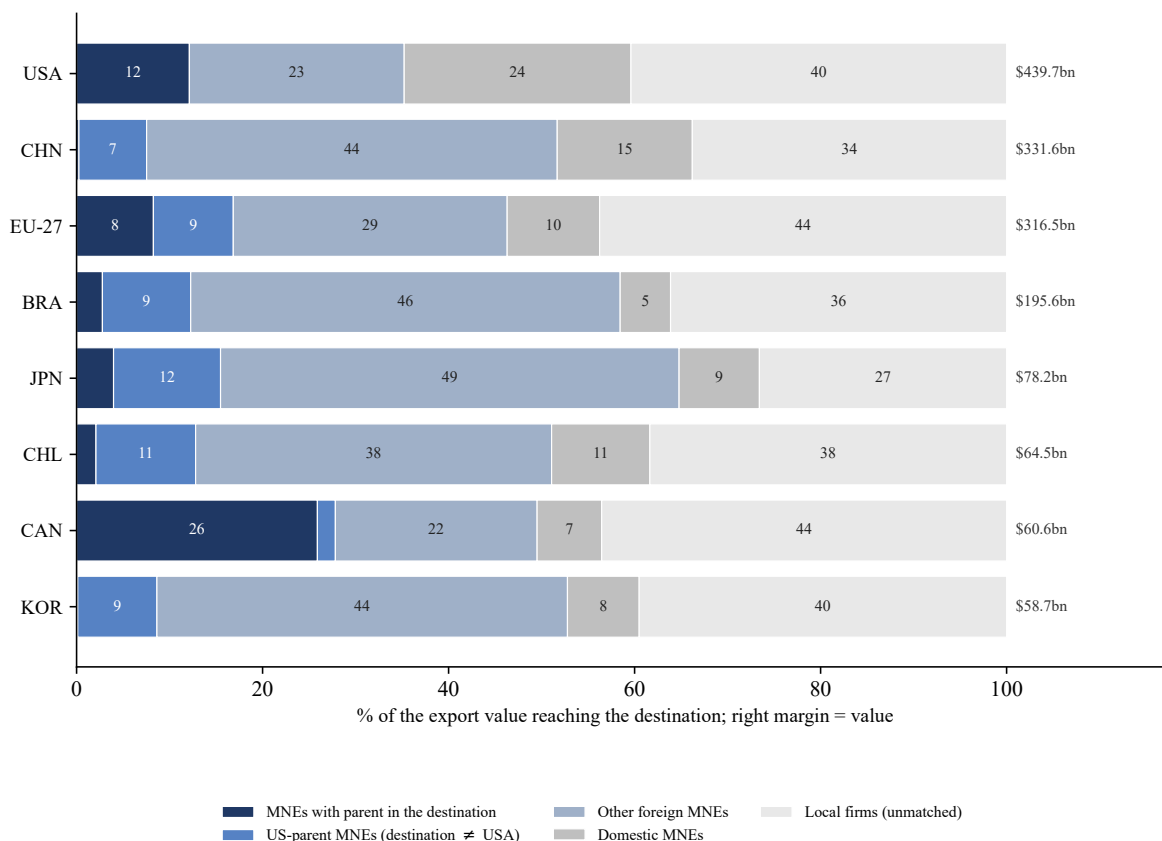
| Sector         | Exports to the USA (\$bn) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|----------------|---------------------------|--------------------|------------------------|-------------------|-----------------|
| All goods      | 439.7                     | 12.1               | 23.1                   | 24.4              | 40.4            |
| Agriculture    | 117.2                     | 3.9                | 28.9                   | 9.2               | 58.0            |
| Mining & fuels | 165.3                     | 6.1                | 15.3                   | 52.0              | 26.6            |
| Manufacturing  | 155.6                     | 24.6               | 26.6                   | 6.7               | 42.1            |
| Rest           | 1.7                       | 24.5               | 53.9                   | 4.5               | 17.1            |

Exports from the nine origins to the United States by sector, and the share of that value moved by each type of exporter. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched.

**Table 19:** Exports to the United States by origin, all goods: who carries them (numbers behind the left panel)

| Origin | Exports to the USA (\$bn) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|--------|---------------------------|--------------------|------------------------|-------------------|-----------------|
| SLV    | 30.2                      | 40.2               | 8.7                    | 0.0               | 51.1            |
| CRI    | 40.1                      | 37.2               | 25.3                   | 4.5               | 32.9            |
| DOM    | 33.0                      | 24.7               | 44.6                   | 1.6               | 29.1            |
| CHL    | 63.4                      | 6.1                | 26.8                   | 15.4              | 51.7            |
| COL    | 166.2                     | 5.7                | 14.7                   | 54.4              | 25.2            |
| PRY    | 1.4                       | 5.6                | 29.9                   | 0.3               | 64.2            |
| ARG    | 37.7                      | 5.0                | 35.7                   | 8.8               | 50.6            |
| PER    | 63.9                      | 4.4                | 25.4                   | 1.9               | 68.2            |
| URY    | 3.9                       | 0.3                | 64.4                   | 5.0               | 30.3            |
| All    | 439.7                     | 12.1               | 23.1                   | 24.4              | 40.4            |

All goods. Exports from each origin to the United States and the share of that value moved by each type of exporter; origins sorted by the US-parent share. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched.

**Figure 22:** The largest destination markets (EU-27 pooled): share of the value reaching each one moved by MNEs with a parent in the destination, US-parent MNEs, other foreign MNEs, domestic MNEs and local firms.

**Table 20:** The largest destination markets: who carries the exports that reach them (%)

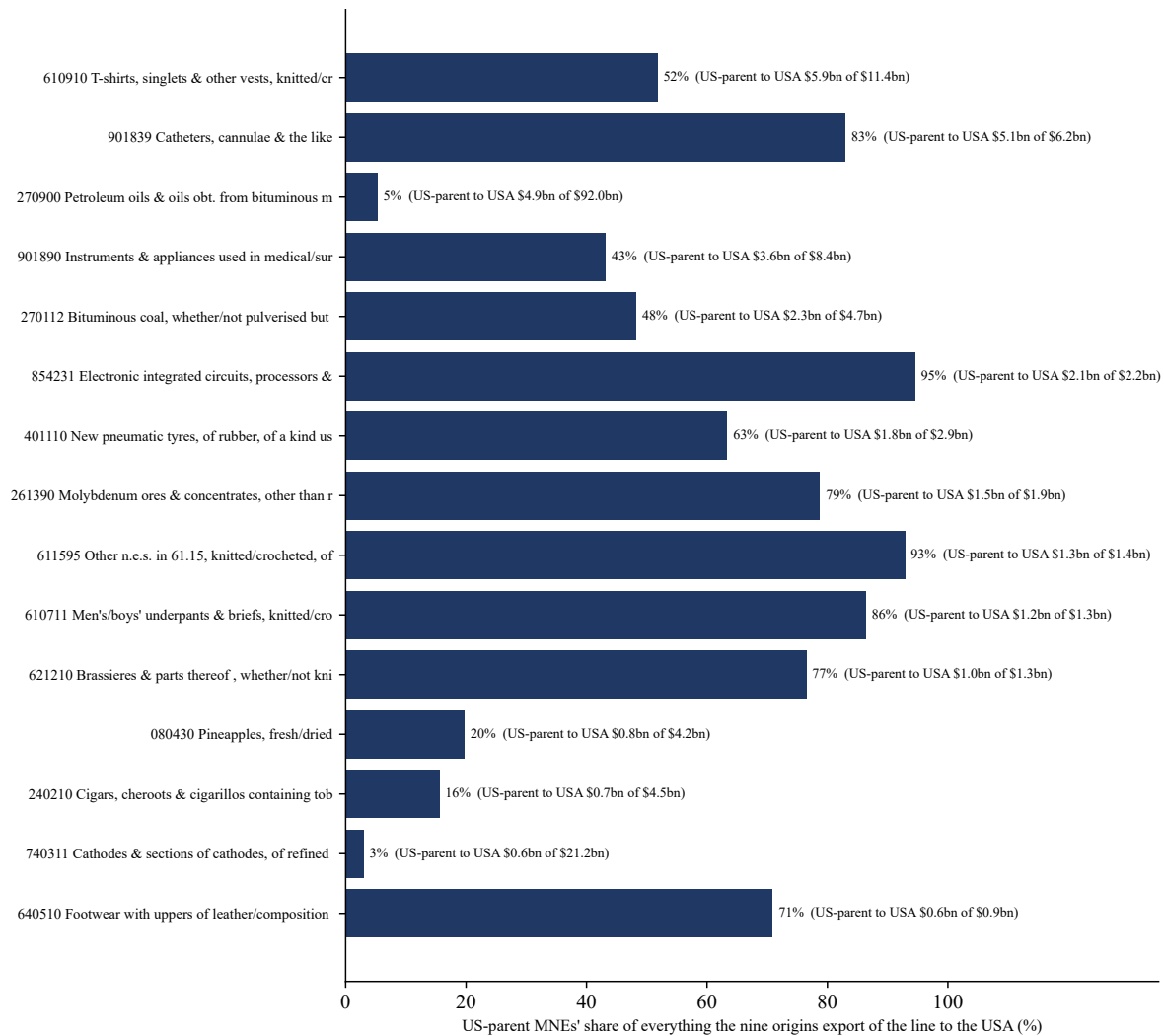
| Destination | Exports (\$bn) | MNEs with parent in the destination (%) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|-------------|----------------|-----------------------------------------|--------------------|------------------------|-------------------|-----------------|
| USA         | 439.7          | 12.1                                    | –                  | 23.1                   | 24.4              | 40.4            |
| CHN         | 331.6          | 0.3                                     | 7.3                | 44.2                   | 14.5              | 33.8            |
| EU-27       | 316.5          | 8.3                                     | 8.6                | 29.5                   | 9.9               | 43.8            |
| BRA         | 195.6          | 2.8                                     | 9.5                | 46.2                   | 5.4               | 36.1            |
| JPN         | 78.2           | 4.0                                     | 11.5               | 49.3                   | 8.6               | 26.6            |
| CHL         | 64.5           | 2.1                                     | 10.7               | 38.3                   | 10.6              | 38.4            |
| CAN         | 60.6           | 25.9                                    | 1.9                | 21.7                   | 6.9               | 43.5            |
| KOR         | 58.7           | 0.1                                     | 8.5                | 44.1                   | 7.7               | 39.5            |

All goods. The 8 largest destination markets (EU-27 pooled) and the share of the value reaching each one moved by each type of exporter. 'Parent in the destination': the exporter's ultimate parent is in the destination country (for the EU-27, in any member state); for the USA this group is the US-parent MNEs and the US-parent column is left blank. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched.

**Table 21:** What US-parent MNEs ship to the United States: top-15 HS6 lines, and the US-parent share of everything LAC exports of the line to the USA

| HS6       | Description                                      | US-parent exports to the USA (\$bn) | % of all US-parent exports to the USA | % of US-parent exports of the line going to the USA | All LAC exports of the line to the USA (\$bn) | US-parent share of LAC exports of the line to the USA (%) |
|-----------|--------------------------------------------------|-------------------------------------|---------------------------------------|-----------------------------------------------------|-----------------------------------------------|-----------------------------------------------------------|
| 610910    | T-shirts, singlets & other vests, knitted/cro... | 5.90                                | 11.1                                  | 84                                                  | 11.39                                         | 52                                                        |
| 901839    | Catheters, cannulae & the like                   | 5.14                                | 9.6                                   | 73                                                  | 6.19                                          | 83                                                        |
| 270900    | Petroleum oils & oils obt. from bituminous mi... | 4.88                                | 9.1                                   | 24                                                  | 92.00                                         | 5                                                         |
| 901890    | Instruments & appliances used in medical/surg... | 3.64                                | 6.8                                   | 69                                                  | 8.42                                          | 43                                                        |
| 270112    | Bituminous coal, whether/not pulverised but n... | 2.27                                | 4.3                                   | 9                                                   | 4.70                                          | 48                                                        |
| 854231    | Electronic integrated circuits, processors & ... | 2.08                                | 3.9                                   | 28                                                  | 2.20                                          | 95                                                        |
| 401110    | New pneumatic tyres, of rubber, of a kind use... | 1.82                                | 3.4                                   | 93                                                  | 2.88                                          | 63                                                        |
| 261390    | Molybdenum ores & concentrates, other than ro... | 1.48                                | 2.8                                   | 99                                                  | 1.88                                          | 79                                                        |
| 611595    | Other n.e.s. in 61.15, knitted/crocheted, of ... | 1.26                                | 2.4                                   | 100                                                 | 1.35                                          | 93                                                        |
| 610711    | Men's/boys' underpants & briefs, knitted/croc... | 1.16                                | 2.2                                   | 87                                                  | 1.34                                          | 86                                                        |
| 621210    | Brassieres & parts thereof, whether/not knit...  | 0.98                                | 1.8                                   | 98                                                  | 1.28                                          | 77                                                        |
| 080430    | Pineapples, fresh/dried                          | 0.83                                | 1.6                                   | 66                                                  | 4.19                                          | 20                                                        |
| 240210    | Cigars, cheroots & cigarillos containing toba... | 0.71                                | 1.3                                   | 100                                                 | 4.53                                          | 16                                                        |
| 740311    | Cathodes & sections of cathodes, of refined c... | 0.63                                | 1.2                                   | 24                                                  | 21.21                                         | 3                                                         |
| 640510    | Footwear with uppers of leather/composition l... | 0.63                                | 1.2                                   | 70                                                  | 0.88                                          | 71                                                        |
| All lines |                                                  | 53.4                                | 100.0                                 | 22                                                  | 439.7                                         | 12                                                        |

Top 15 HS6 lines by the value US-parent multinationals export from the nine origins to the United States. Column 5 = of what US-parent MNEs export of the line from LAC (all destinations), the share shipped to the USA; column 7 = of what all firms in the nine origins export of the line to the USA, the share moved by US-parent MNEs. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched.



**Figure 23:** US-parent MNEs' share of LAC exports to the United States, for the 15 lines they ship home in the largest value.

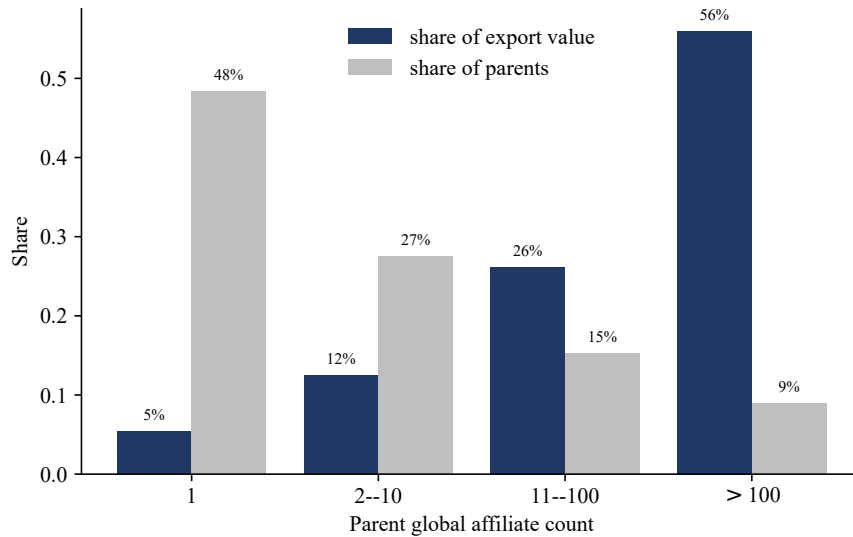
**Table 22:** The 15 largest LAC export lines to the United States: who carries them (%)

| HS6    | Description                                      | LAC exports to the USA (\$bn) | US-parent MNEs (%) | Other foreign MNEs (%) | Domestic MNEs (%) | Local firms (%) |
|--------|--------------------------------------------------|-------------------------------|--------------------|------------------------|-------------------|-----------------|
| 270900 | Petroleum oils & oils obt. from bituminous mi... | 92.00                         | 5                  | 10                     | 81                | 4               |
| 710812 | Gold (incl. gold plated with platinum), in un... | 29.61                         | 1                  | 16                     | 9                 | 74              |
| 740311 | Cathodes & sections of cathodes, of refined c... | 21.21                         | 3                  | 33                     | 4                 | 60              |
| 090111 | Coffee, not roasted, not decaffeinated           | 16.36                         | 2                  | 37                     | 8                 | 53              |
| 271019 | Petroleum oils & oils obtained from bituminou... | 12.81                         | 0                  | 14                     | 51                | 35              |
| 610910 | T-shirts, singlets & other vests, knitted/cro... | 11.39                         | 52                 | 8                      | 0                 | 40              |
| 901890 | Instruments & appliances used in medical/surg... | 8.42                          | 43                 | 48                     | 0                 | 9               |
| 060319 | Cut flowers & flower buds of a kind suit. for... | 7.33                          | 1                  | 20                     | 19                | 60              |
| 271011 | Light petroleum oils & preparations              | 6.95                          | 2                  | 17                     | 5                 | 76              |
| 901839 | Catheters, cannulae & the like                   | 6.19                          | 83                 | 8                      | 0                 | 9               |
| 030441 | Fish fillets&other fish meat (excl. of 0304.1... | 5.25                          | 2                  | 18                     | 24                | 56              |
| 080610 | Grapes, fresh                                    | 4.76                          | 3                  | 29                     | 10                | 58              |
| 270112 | Bituminous coal, whether/not pulverised but n... | 4.70                          | 48                 | 32                     | 1                 | 19              |
| 240210 | Cigars, cheroots & cigarillos containing toba... | 4.53                          | 16                 | 47                     | 0                 | 37              |
| 080390 | Bananas, fresh or dried (excl. plantains) [HS... | 4.52                          | 13                 | 27                     | 1                 | 58              |

Top 15 HS6 lines by the value the nine origins export to the United States, and the share of each line moved by each type of exporter. Pooled 2006–2022, nine LAC origins (Ecuador excluded), value-weighted. US-parent MNE = matched exporter whose recorded ultimate parent is in the United States; other foreign MNE = every other matched exporter with a parent abroad or with no recorded parent; domestic MNE = parent in the exporting country; local = unmatched.

## 5 Fact 4 — A small set of large multinational groups accounts for the bulk of exports

### 5.1 Original exhibits

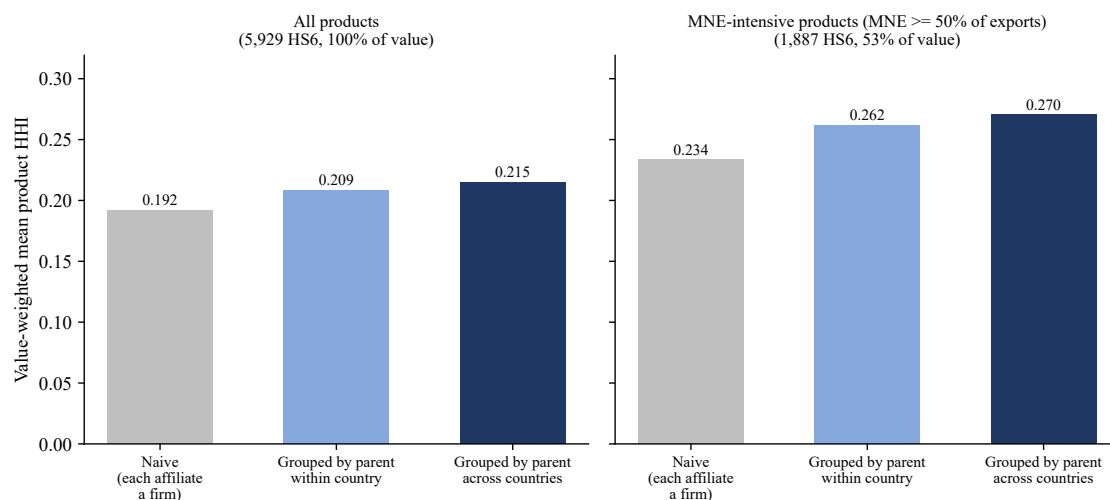


**Figure 24:** Figure 5 of the note, reproduced: foreign-MNE export value and parent counts by global affiliate-network size.

**Table 23:** Numbers behind Figure 5

| Global affiliate count            | Parents | Share of parents (%) | Export value (\$bn) | Share of value (%) |
|-----------------------------------|---------|----------------------|---------------------|--------------------|
| 1                                 | 5,982   | 48.4                 | 54.3                | 5.4                |
| 2-10                              | 3,397   | 27.5                 | 124.4               | 12.4               |
| 11-100                            | 1,893   | 15.3                 | 261.7               | 26.2               |
| >100                              | 1,099   | 8.9                  | 560.2               | 56.0               |
| All parents with a network record | 12,371  | 100                  | 1,000.6             | 100                |

Foreign-MNE parents (parent name key) exporting from the nine origins; 98% of parents and 97% of foreign-MNE export value have a matched global-network record (Orbis  $\cup$  D&B roster, src/16). Network size = number of distinct worldwide affiliates.



**Figure 25:** Figure 6 of the note, reproduced: product-level export concentration, naive firm count vs grouping affiliates by parent.

**Table 24:** Numbers behind Figure 6: HHI, top-exporter share, effective number of exporters

| Sample                                                                                        | Measure                               | Naive | Within country | Across countries |
|-----------------------------------------------------------------------------------------------|---------------------------------------|-------|----------------|------------------|
| <i>All products: 5,929 HS6, 100% of value</i>                                                 |                                       |       |                |                  |
|                                                                                               | Mean HHI                              | 0.192 | 0.209          | 0.215            |
|                                                                                               | Mean top-exporter share               | 31.0% | 32.5%          | 33.7%            |
|                                                                                               | Effective number of exporters (1/HHI) | 12.5  | 11.7           | 10.4             |
| <i>MNE-intensive products (MNE <math>\geq</math> 50% of exports): 1,887 HS6, 53% of value</i> |                                       |       |                |                  |
|                                                                                               | Mean HHI                              | 0.234 | 0.262          | 0.270            |
|                                                                                               | Mean top-exporter share               | 36.0% | 38.7%          | 39.8%            |
|                                                                                               | Effective number of exporters (1/HHI) | 8.9   | 8.0            | 7.4              |

Value-weighted means across HS6 products (nine origins, Ecuador excluded). Naive: each exporting affiliate (Tax ID) is a firm; then affiliates of the same global parent are grouped within a country and across countries.

## 5.2 Variant: groups versus affiliates inside the market cells (item 1e, “MNE cantidad”)

**Table 25:** Affiliates per group in the origin–destination–product–year cells

| Origin-destination-product-year cells with $\geq 1$ MNE   |           |
|-----------------------------------------------------------|-----------|
| Cells                                                     | 1,008,755 |
| MNE affiliates per cell (mean)                            | 1.91      |
| MNE groups per cell (mean)                                | 1.90      |
| Affiliates per group, pooled                              | 1.01      |
| Affiliates per group, value-weighted mean                 | 1.08      |
| Cells where one group exports through $> 1$ affiliate (%) | 1.0       |

Group = ultimate parent (Orbis GUO id or D&B global ultimate; parent name when the id is missing).

**Table 26:** Fact-5 regressions with ln(# MNE groups) in place of ln(# MNE firms)

|                                           | OxYr + DxYr + P FE    |                       |                       | ODP + ODY FE          |                       |                       |
|-------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                                           | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   |
| <i>Panel A: all exports (ln)</i>          |                       |                       |                       |                       |                       |                       |
| ln(# MNE firms)                           | 2.0685***<br>(0.0368) |                       |                       | 0.9414***<br>(0.0206) |                       |                       |
| ln(# MNE groups)                          |                       | 2.0728***<br>(0.0371) | 2.0617***<br>(0.0365) |                       | 0.9416***<br>(0.0208) | 0.9421***<br>(0.0208) |
| ln(affiliates per group)                  |                       |                       | 2.7723***<br>(0.2413) |                       |                       | 0.8692***<br>(0.0648) |
| <i>Panel B: non-MNE exports (ln)</i>      |                       |                       |                       |                       |                       |                       |
| ln(# MNE firms)                           | 1.3406***<br>(0.0319) |                       |                       | 0.2389***<br>(0.0178) |                       |                       |
| ln(# MNE groups)                          |                       | 1.3471***<br>(0.0319) | 1.3427***<br>(0.0317) |                       | 0.2400***<br>(0.0177) | 0.2400***<br>(0.0177) |
| ln(affiliates per group)                  |                       |                       | 1.1055***<br>(0.2963) |                       |                       | 0.0605<br>(0.0985)    |
| Product FE                                | ✓                     | ✓                     | ✓                     |                       |                       |                       |
| Origin $\times$ year FE                   | ✓                     | ✓                     | ✓                     |                       |                       |                       |
| Destination $\times$ year FE              | ✓                     | ✓                     | ✓                     |                       |                       |                       |
| Origin $\times$ dest. $\times$ product FE |                       |                       |                       | ✓                     | ✓                     | ✓                     |
| Origin $\times$ dest. $\times$ year FE    |                       |                       |                       | ✓                     | ✓                     | ✓                     |
| Observations (all exports)                | 1,008,185             | 1,008,185             | 1,008,185             | 883,545               | 883,545               | 883,545               |
| Observations (non-MNE exports)            | 475,514               | 475,514               | 475,514               | 429,882               | 429,882               | 429,882               |

Cells with at least one MNE; dep. var. ln exports. ‘MNE firms’ counts matched exporting affiliates; ‘MNE groups’ counts distinct ultimate parents. SE clustered at origin-destination. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

## 6 Fact 5 — Greater multinational presence is associated with higher trade volumes

### 6.1 Original exhibit

**Table 27:** Table 1 of the note, reproduced: multinational presence and trade volume (intensive and extensive margins)

|                                      | Intensive: ln(# MNE firms) |                       |                       | Extensive: any MNE present |                       |                       |
|--------------------------------------|----------------------------|-----------------------|-----------------------|----------------------------|-----------------------|-----------------------|
|                                      | (1)                        | (2)                   | (3)                   | (1)                        | (2)                   | (3)                   |
| <i>Panel A: all exports (ln)</i>     |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                      | 2.0823***<br>(0.0363)      | 2.0685***<br>(0.0368) | 0.9414***<br>(0.0206) |                            |                       |                       |
| Any MNE present                      |                            |                       |                       | 1.4622***<br>(0.0394)      | 1.4591***<br>(0.0388) | 0.6818***<br>(0.0165) |
| <i>Panel B: non-MNE exports (ln)</i> |                            |                       |                       |                            |                       |                       |
| ln(# MNE firms)                      | 1.3536***<br>(0.0318)      | 1.3406***<br>(0.0319) | 0.2389***<br>(0.0178) |                            |                       |                       |
| Any MNE present                      |                            |                       |                       | 1.4168***<br>(0.0371)      | 1.3996***<br>(0.0367) | 0.1311***<br>(0.0102) |
| Origin FE                            | ✓                          |                       |                       | ✓                          |                       |                       |
| Destination FE                       | ✓                          |                       |                       | ✓                          |                       |                       |
| Product FE                           | ✓                          | ✓                     |                       | ✓                          | ✓                     |                       |
| Year FE                              | ✓                          |                       |                       | ✓                          |                       |                       |
| Origin × year FE                     |                            | ✓                     |                       |                            | ✓                     |                       |
| Destination × year FE                |                            | ✓                     |                       |                            | ✓                     |                       |
| Origin × dest. × product FE          |                            |                       | ✓                     |                            |                       | ✓                     |
| Origin × dest. × year FE             |                            |                       | ✓                     |                            |                       | ✓                     |
| Observations (all exports)           | 1,008,472                  | 1,008,185             | 883,545               | 1,829,852                  | 1,829,626             | 1,633,319             |
| Observations (non-MNE exports)       | 475,856                    | 475,514               | 429,882               | 1,297,322                  | 1,297,111             | 1,137,550             |

Origin-destination-product-year cells; dep. var. ln exports. Intensive margin: ln number of MNE firms (cells with  $\geq 1$  MNE); extensive margin: indicator for any MNE present. Reproduction of the note's Table 1 on the current base. SE clustered at origin-destination. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

### 6.2 Variant: foreign vs domestic MNEs; foreign MNEs present through headquarters vs not (item 1e)

**Table 28:** MNE export value by origin: domestic share, and foreign-MNE value by the group's presence at the destination

| Origin | MNE value (\$bn) | of which domestic (%) | Foreign-MNE value (\$bn) | % of foreign-MNE value: group present at destination |                                                 |
|--------|------------------|-----------------------|--------------------------|------------------------------------------------------|-------------------------------------------------|
|        |                  |                       |                          | through headquarters                                 | not through HQ (of which via another affiliate) |
| ARG    | 337.1            | 13.3                  | 292.3                    | 6.5                                                  | 93.5 (41.8)                                     |
| CHL    | 294.4            | 22.1                  | 229.3                    | 4.3                                                  | 95.7 (56.7)                                     |
| COL    | 393.0            | 52.5                  | 186.5                    | 6.5                                                  | 93.5 (34.2)                                     |
| CRI    | 66.3             | 5.7                   | 62.5                     | 25.5                                                 | 74.5 (33.8)                                     |
| DOM    | 48.0             | 1.8                   | 47.1                     | 29.0                                                 | 71.0 (21.6)                                     |
| PER    | 193.0            | 5.0                   | 183.3                    | 3.9                                                  | 96.1 (28.8)                                     |
| PRY    | 40.7             | 1.8                   | 40.0                     | 2.1                                                  | 97.9 (65.5)                                     |
| SLV    | 31.2             | 0.6                   | 31.0                     | 40.7                                                 | 59.3 (28.1)                                     |
| URY    | 44.2             | 15.9                  | 37.2                     | 5.2                                                  | 94.8 (37.7)                                     |
| All    | 1,447.8          | 23.4                  | 1,109.2                  | 8.4                                                  | 91.6 (40.5)                                     |

'Through headquarters': the affiliate ships to its parent's country. 'Not through HQ': every other destination; in parentheses the part where the group has another affiliate in the destination (Orbis/D&B roster). Pooled 2006–2022.

**Table 29:** Intensive margin:  $\ln(\# \text{ MNE firms})$  decomposed into foreign / domestic and foreign through HQ / not through HQ

|                                                       | All MNEs              |                       | Foreign / domestic    |                       | Foreign through HQ / not through HQ / domestic |                       |
|-------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------------------------------|-----------------------|
|                                                       | (1)                   | (2)                   | (3)                   | (4)                   | (5)                                            | (6)                   |
| <i>Panel A: all exports (ln)</i>                      |                       |                       |                       |                       |                                                |                       |
| $\ln(\# \text{ MNE firms})$                           | 2.0685***<br>(0.0368) | 0.9414***<br>(0.0206) |                       |                       |                                                |                       |
| $\ln(\# \text{ foreign MNEs})$                        |                       |                       | 1.2129***<br>(0.0285) | 0.4506***<br>(0.0124) |                                                |                       |
| $\ln(\# \text{ foreign MNEs present through HQ})$     |                       |                       |                       |                       | 0.3970***<br>(0.0856)                          | 0.1448***<br>(0.0370) |
| $\ln(\# \text{ foreign MNEs not present through HQ})$ |                       |                       |                       |                       | 0.9465***<br>(0.0536)                          | 0.3133***<br>(0.0382) |
| $\ln(\# \text{ domestic MNEs})$                       |                       |                       | 0.4382***<br>(0.0259) | 0.2214***<br>(0.0151) | 0.3770***<br>(0.0514)                          | 0.1555***<br>(0.0315) |
| <i>Panel B: non-MNE exports (ln)</i>                  |                       |                       |                       |                       |                                                |                       |
| $\ln(\# \text{ MNE firms})$                           | 1.3406***<br>(0.0319) | 0.2389***<br>(0.0178) |                       |                       |                                                |                       |
| $\ln(\# \text{ foreign MNEs})$                        |                       |                       | 1.1331***<br>(0.0375) | 0.2413***<br>(0.0217) |                                                |                       |
| $\ln(\# \text{ foreign MNEs present through HQ})$     |                       |                       |                       |                       | 0.2934***<br>(0.0882)                          | 0.1945***<br>(0.0527) |
| $\ln(\# \text{ foreign MNEs not present through HQ})$ |                       |                       |                       |                       | 1.0330***<br>(0.0607)                          | 0.1951***<br>(0.0648) |
| $\ln(\# \text{ domestic MNEs})$                       |                       |                       | 0.4707***<br>(0.0341) | 0.1250***<br>(0.0218) | 0.4142***<br>(0.0633)                          | 0.0946*<br>(0.0552)   |
| Product FE                                            | ✓                     |                       | ✓                     |                       | ✓                                              |                       |
| Origin $\times$ year FE                               | ✓                     |                       | ✓                     |                       | ✓                                              |                       |
| Destination $\times$ year FE                          | ✓                     |                       | ✓                     |                       | ✓                                              |                       |
| Origin $\times$ dest. $\times$ product FE             |                       | ✓                     |                       | ✓                     |                                                | ✓                     |
| Origin $\times$ dest. $\times$ year FE                |                       | ✓                     |                       | ✓                     |                                                | ✓                     |
| Observations (all exports)                            | 1,008,185             | 883,545               | 113,422               | 98,223                | 9,197                                          | 7,524                 |
| Observations (non-MNE exports)                        | 475,514               | 429,882               | 93,010                | 81,563                | 8,780                                          | 7,255                 |

Origin-destination-product-year cells; dep. var.  $\ln$  exports. Intensive margin: plain logs of the counts, so each column uses the cells where every count entering it is positive (no  $\ln(1+x)$ ). (1)–(2) all MNEs; (3)–(4) foreign vs domestic; (5)–(6) foreign split by whether the group is present at the destination through its headquarters (the parent's country). SE clustered at origin-destination. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

**Table 30:** Extensive margin: presence dummies decomposed into foreign / domestic and foreign through HQ / not through HQ

|                                        | All MNEs              |                       | Foreign / domestic    |                       | Foreign through HQ / not through HQ / domestic |                       |
|----------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------------------------------|-----------------------|
|                                        | (1)                   | (2)                   | (3)                   | (4)                   | (5)                                            | (6)                   |
| <i>Panel A: all exports (ln)</i>       |                       |                       |                       |                       |                                                |                       |
| Any MNE present                        | 1.4591***<br>(0.0388) | 0.6818***<br>(0.0165) |                       |                       |                                                |                       |
| Any foreign MNE                        |                       |                       | 1.4990***<br>(0.0339) | 0.6986***<br>(0.0158) |                                                |                       |
| Any foreign MNE present through HQ     |                       |                       |                       |                       | 1.3873***<br>(0.0951)                          | 0.5909***<br>(0.0431) |
| Any foreign MNE not present through HQ |                       |                       |                       |                       | 1.5093***<br>(0.0319)                          | 0.6976***<br>(0.0146) |
| Any domestic MNE                       |                       |                       | 1.4670***<br>(0.0416) | 0.4917***<br>(0.0165) | 1.4334***<br>(0.0388)                          | 0.4909***<br>(0.0166) |
| <i>Panel B: non-MNE exports (ln)</i>   |                       |                       |                       |                       |                                                |                       |
| Any MNE present                        | 1.3996***<br>(0.0367) | 0.1311***<br>(0.0102) |                       |                       |                                                |                       |
| Any foreign MNE                        |                       |                       | 1.2983***<br>(0.0346) | 0.1356***<br>(0.0106) |                                                |                       |
| Any foreign MNE present through HQ     |                       |                       |                       |                       | 1.0183***<br>(0.0900)                          | 0.1265***<br>(0.0194) |
| Any foreign MNE not present through HQ |                       |                       |                       |                       | 1.2782***<br>(0.0326)                          | 0.1386***<br>(0.0107) |
| Any domestic MNE                       |                       |                       | 1.1316***<br>(0.0428) | 0.1370***<br>(0.0138) | 1.0959***<br>(0.0403)                          | 0.1367***<br>(0.0138) |
| Product FE                             | ✓                     |                       | ✓                     |                       | ✓                                              |                       |
| Origin × year FE                       | ✓                     |                       | ✓                     |                       | ✓                                              |                       |
| Destination × year FE                  | ✓                     |                       | ✓                     |                       | ✓                                              |                       |
| Origin × dest. × product FE            |                       | ✓                     |                       | ✓                     |                                                | ✓                     |
| Origin × dest. × year FE               |                       | ✓                     |                       | ✓                     |                                                | ✓                     |
| Observations (all exports)             | 1,829,626             | 1,633,319             | 1,829,626             | 1,633,319             | 1,829,626                                      | 1,633,319             |
| Observations (non-MNE exports)         | 1,297,111             | 1,137,550             | 1,297,111             | 1,137,550             | 1,297,111                                      | 1,137,550             |

Origin-destination-product-year cells; dep. var. ln exports. Extensive margin: indicators for the presence of at least one MNE of each type in the cell; all cells. SE clustered at origin-destination. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

### 6.3 Summary across sectors (revision 6)

**Table 31:** Fact 5 in one table:  $\ln(\# \text{ foreign MNEs})$  and  $\ln(\# \text{ domestic MNEs})$ , all goods and by sector

|                                                                                                            | All goods             | Agriculture           | Mining & fuels        | Manufacturing         |
|------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <i>Panel A: all exports of the cell (ln)</i>                                                               |                       |                       |                       |                       |
| Origin $\times$ year, destination $\times$ year and product FE (column (3) of the full table)              |                       |                       |                       |                       |
| $\ln(\# \text{ foreign MNEs})$                                                                             | 1.2129***<br>(0.0285) | 1.2954***<br>(0.0275) | 1.1301***<br>(0.1799) | 1.1644***<br>(0.0358) |
| $\ln(\# \text{ domestic MNEs})$                                                                            | 0.4382***<br>(0.0259) | 0.5233***<br>(0.0266) | 0.3172<br>(0.2062)    | 0.3833***<br>(0.0319) |
| Observations                                                                                               | 113,422               | 25,437                | 1,417                 | 85,453                |
| Origin $\times$ destination $\times$ product and origin $\times$ destination $\times$ year FE (column (4)) |                       |                       |                       |                       |
| $\ln(\# \text{ foreign MNEs})$                                                                             | 0.4506***<br>(0.0124) | 0.4625***<br>(0.0188) | 0.2479<br>(0.1778)    | 0.4435***<br>(0.0148) |
| $\ln(\# \text{ domestic MNEs})$                                                                            | 0.2214***<br>(0.0151) | 0.2831***<br>(0.0234) | 0.3216*<br>(0.1639)   | 0.1904***<br>(0.0175) |
| Observations                                                                                               | 98,223                | 22,002                | 1,027                 | 73,318                |
| <i>Panel B: exports of the cell's non-MNE firms (ln)</i>                                                   |                       |                       |                       |                       |
| Origin $\times$ year, destination $\times$ year and product FE (column (3) of the full table)              |                       |                       |                       |                       |
| $\ln(\# \text{ foreign MNEs})$                                                                             | 1.1331***<br>(0.0375) | 1.2191***<br>(0.0363) | 0.9420***<br>(0.2641) | 1.0546***<br>(0.0493) |
| $\ln(\# \text{ domestic MNEs})$                                                                            | 0.4707***<br>(0.0341) | 0.5676***<br>(0.0339) | -0.6594<br>(0.4830)   | 0.4194***<br>(0.0419) |
| Observations                                                                                               | 93,010                | 22,503                | 995                   | 68,534                |
| Origin $\times$ destination $\times$ product and origin $\times$ destination $\times$ year FE (column (4)) |                       |                       |                       |                       |
| $\ln(\# \text{ foreign MNEs})$                                                                             | 0.2413***<br>(0.0217) | 0.3602***<br>(0.0292) | -0.1512<br>(0.3320)   | 0.1865***<br>(0.0250) |
| $\ln(\# \text{ domestic MNEs})$                                                                            | 0.1250***<br>(0.0218) | 0.2461***<br>(0.0341) | -0.7273<br>(0.6536)   | 0.0742***<br>(0.0238) |
| Observations                                                                                               | 81,563                | 19,535                | 726                   | 59,772                |

Fact 5 in one table: the coefficients of  $\ln(\# \text{ foreign MNEs})$  and  $\ln(\# \text{ domestic MNEs})$  from the intensive-margin decomposition (columns (3) and (4) of the full table in each scope), all goods and within each sector. Origin–destination–product–year cells with at least one foreign and one domestic MNE; plain logs; dep. var.  $\ln$  exports of the cell (Panel A) or of its non-MNE exporters (Panel B). A blank cell was not estimated (too few observations). SE clustered at origin–destination in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

## 7 Fact 6 — Distance is a weaker barrier to trade for multinational corporations

### 7.1 Original exhibit

**Table 32:** Table 2 of the note, reproduced: distance and firm exports, multinationals split by presence at the destination

| Dep. var.: ln exports          | (1)                    | (2)                    | (3)                    | (4)                    |
|--------------------------------|------------------------|------------------------|------------------------|------------------------|
| ln distance                    | -0.1639***<br>(0.0454) | -0.1848***<br>(0.0460) | -0.1632***<br>(0.0452) | -0.1838***<br>(0.0459) |
| ln distance × MNE              | 0.0461***<br>(0.0038)  | 0.0458***<br>(0.0038)  |                        |                        |
| ln distance × MNE, present     |                        |                        | 0.0689***<br>(0.0070)  | 0.0682***<br>(0.0070)  |
| ln distance × MNE, not present |                        |                        | 0.0393***<br>(0.0032)  | 0.0391***<br>(0.0032)  |
| Origin FE                      | ✓                      |                        | ✓                      |                        |
| Destination FE                 | ✓                      |                        | ✓                      |                        |
| Year FE                        | ✓                      |                        | ✓                      |                        |
| Origin × year FE               |                        | ✓                      |                        | ✓                      |
| Destination × year FE          |                        | ✓                      |                        | ✓                      |
| Product FE                     | ✓                      | ✓                      | ✓                      | ✓                      |
| Observations                   | 5,081,547              | 5,081,360              | 5,081,547              | 5,081,360              |

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. Reproduction of the note's Table 2 on the current base: multinationals (foreign or domestic) split by presence at the destination (the group has an affiliate there, or the destination is the parent's country). SE clustered at origin-destination. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

### 7.2 Variant: foreign vs domestic MNEs; foreign MNEs present through headquarters vs not (item 1e)

**Table 33:** Distance and firm exports: foreign / domestic, then foreign through HQ / not through HQ

| Dep. var.: ln exports                             | (1)                    | (2)                    | (3)                    | (4)                    |
|---------------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| ln distance                                       | -0.1645***<br>(0.0454) | -0.1853***<br>(0.0460) | -0.1644***<br>(0.0454) | -0.1851***<br>(0.0460) |
| ln distance × foreign MNE                         | 0.0487***<br>(0.0041)  | 0.0483***<br>(0.0040)  |                        |                        |
| ln distance × foreign MNE, present through HQ     |                        |                        | 0.0642***<br>(0.0159)  | 0.0638***<br>(0.0161)  |
| ln distance × foreign MNE, not present through HQ |                        |                        | 0.0477***<br>(0.0035)  | 0.0473***<br>(0.0035)  |
| ln distance × domestic MNE                        | 0.0318***<br>(0.0042)  | 0.0318***<br>(0.0042)  | 0.0317***<br>(0.0042)  | 0.0317***<br>(0.0042)  |
| Origin FE                                         | ✓                      |                        | ✓                      |                        |
| Destination FE                                    | ✓                      |                        | ✓                      |                        |
| Year FE                                           | ✓                      |                        | ✓                      |                        |
| Origin × year FE                                  |                        | ✓                      |                        | ✓                      |
| Destination × year FE                             |                        | ✓                      |                        | ✓                      |
| Product FE                                        | ✓                      | ✓                      | ✓                      | ✓                      |
| Observations                                      | 5,081,547              | 5,081,360              | 5,081,547              | 5,081,360              |

Distance and firm exports. Firm×origin×destination×product×year; non-MNE flows the omitted base. (1)–(2) foreign vs domestic multinationals; (3)–(4) foreign MNEs split by whether the group is present at the destination through its headquarters (the parent's country) or not. SE clustered at origin-destination. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1

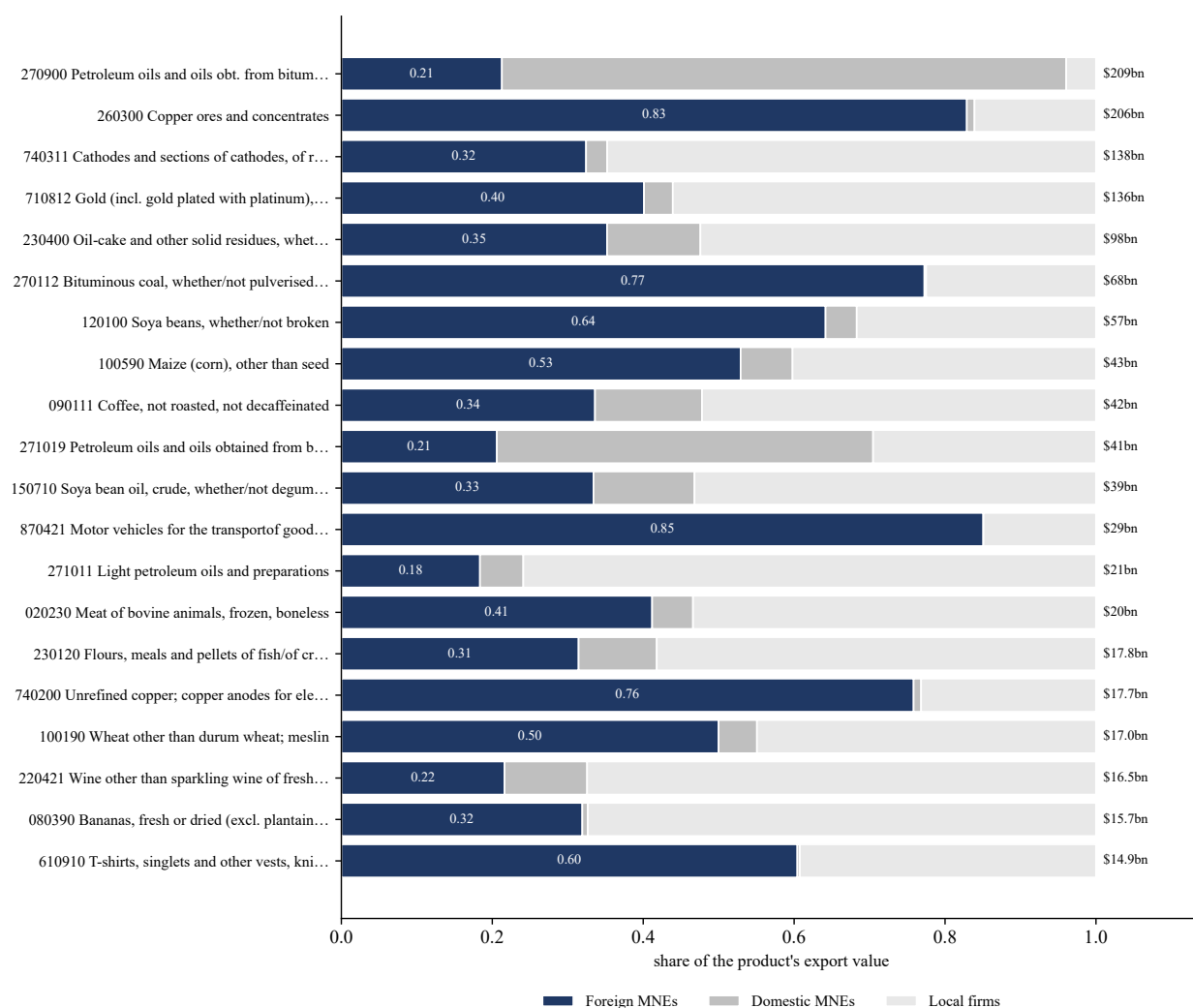
### 7.3 Summary across sectors (revision 6)

**Table 34:** Fact 6 in one table: distance elasticity and its attenuation for foreign and domestic MNEs, all goods and by sector

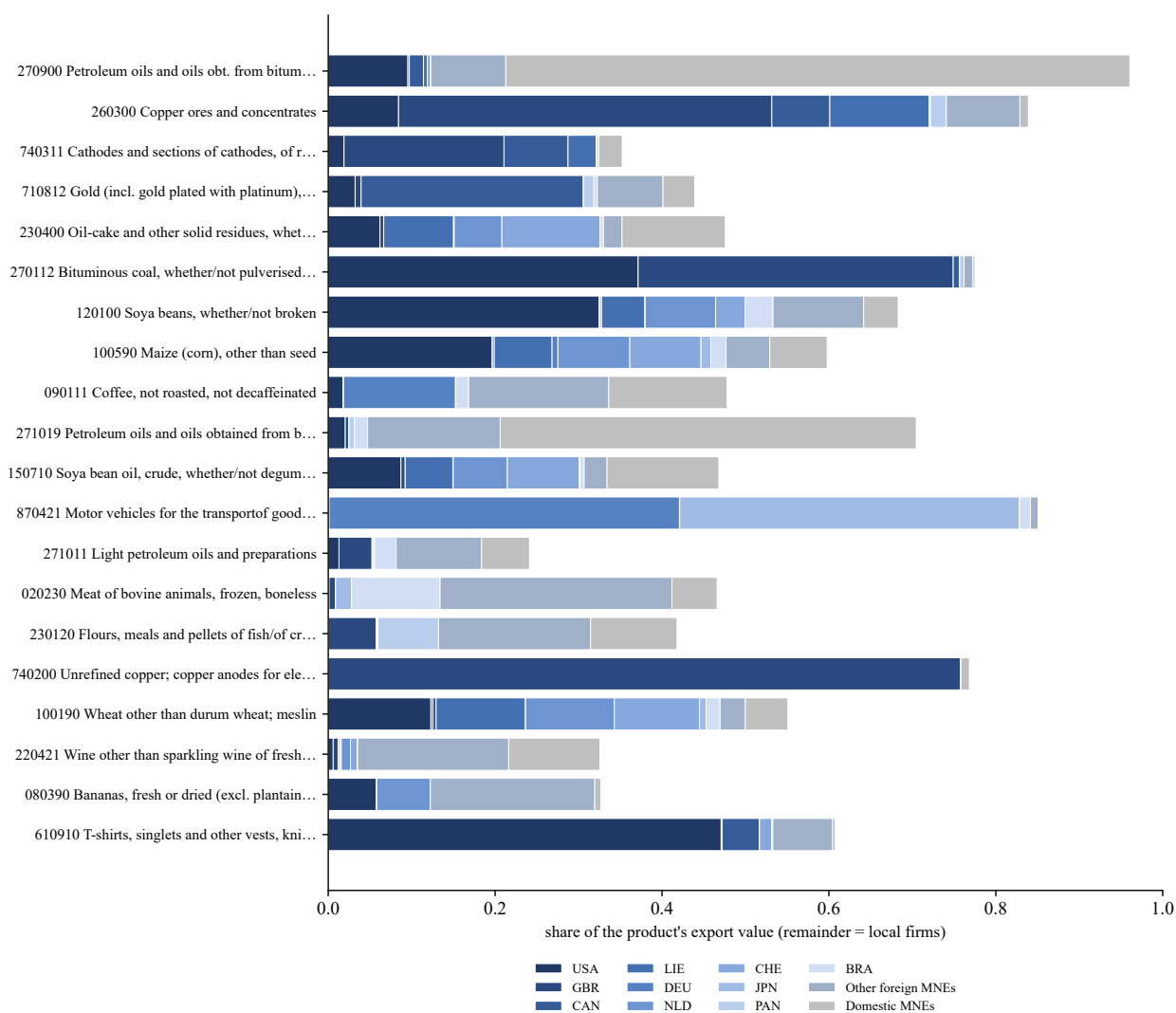
| Dep. var.: ln firm exports                                                  | All goods              | Agriculture            | Mining & fuels        | Manufacturing          |
|-----------------------------------------------------------------------------|------------------------|------------------------|-----------------------|------------------------|
| Origin, destination, year and product FE (column (1) of the full table)     |                        |                        |                       |                        |
| ln distance                                                                 | -0.1645***<br>(0.0454) | -0.2135***<br>(0.0748) | -0.0926<br>(0.0819)   | -0.1510***<br>(0.0454) |
| ln distance $\times$ foreign MNE                                            | 0.0487***<br>(0.0041)  | 0.0366***<br>(0.0034)  | 0.0566***<br>(0.0086) | 0.0515***<br>(0.0048)  |
| ln distance $\times$ domestic MNE                                           | 0.0318***<br>(0.0042)  | 0.0187***<br>(0.0046)  | 0.0706***<br>(0.0180) | 0.0388***<br>(0.0052)  |
| Observations                                                                | 5,081,547              | 1,045,009              | 79,938                | 3,917,960              |
| Origin $\times$ year, destination $\times$ year and product FE (column (2)) |                        |                        |                       |                        |
| ln distance                                                                 | -0.1853***<br>(0.0460) | -0.2285***<br>(0.0711) | -0.1001<br>(0.0828)   | -0.1742***<br>(0.0459) |
| ln distance $\times$ foreign MNE                                            | 0.0483***<br>(0.0040)  | 0.0358***<br>(0.0034)  | 0.0557***<br>(0.0087) | 0.0512***<br>(0.0048)  |
| ln distance $\times$ domestic MNE                                           | 0.0318***<br>(0.0042)  | 0.0182***<br>(0.0046)  | 0.0673***<br>(0.0183) | 0.0390***<br>(0.0052)  |
| Observations                                                                | 5,081,360              | 1,044,787              | 79,612                | 3,917,765              |

Fact 6 in one table: the distance elasticity of firm exports and its attenuation for foreign and for domestic multinationals (columns (1) and (2) of the full table in each scope), all goods and within each sector. Firm  $\times$  origin  $\times$  destination  $\times$  product  $\times$  year; non-MNE flows are the omitted base, so the interaction is the difference from the base elasticity. SE clustered at origin–destination in parentheses. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

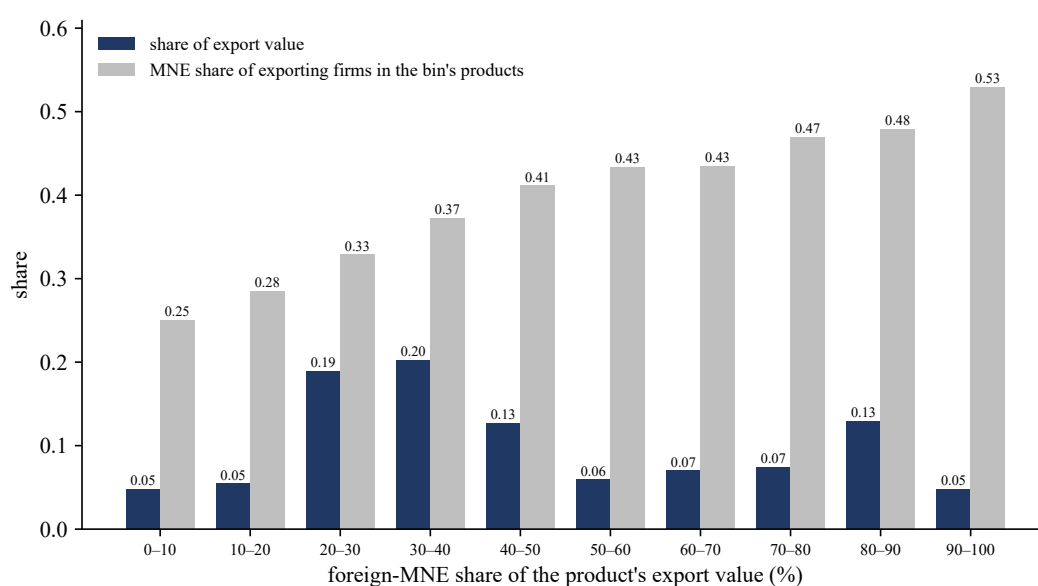
## 8 Products at HS6, with descriptions (item 1f)



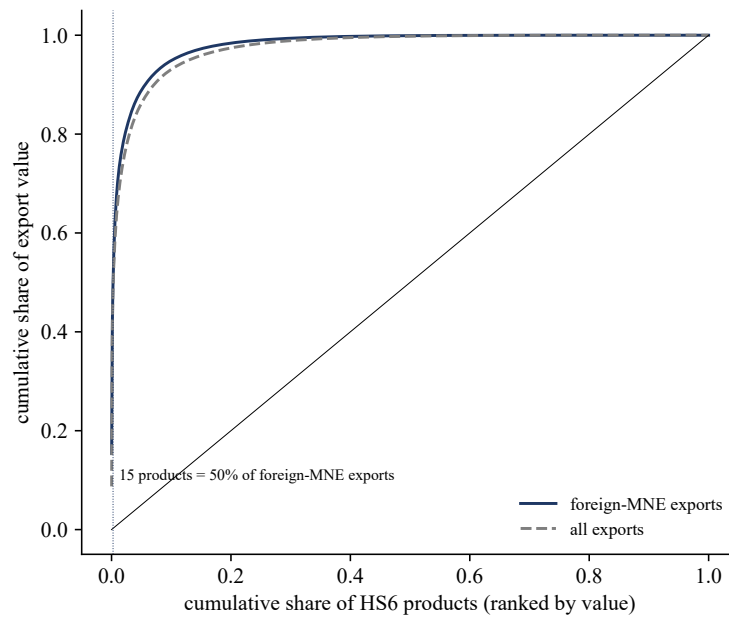
**Figure 26:** Top-20 HS6 products by export value: foreign MNE / domestic MNE / local shares.



**Figure 27:** Top-20 HS6 products: foreign bar split by parent country (remainder = local firms).



**Figure 28:** Distribution of the HS6 foreign-MNE share: share of export value, and MNE share of exporting firms, by bin.



**Figure 29:** Concentration of exports across HS6 products: foreign-MNE exports vs all exports.

**Table 35:** HS sections by export value: MNE shares and leading parent

| HS section                                 | \$bn  | For. | Dom. | Local | Lead parent |
|--------------------------------------------|-------|------|------|-------|-------------|
| V Mineral products                         | 655.5 | 51   | 29   | 19    | GBR (0.41)  |
| II Vegetable products                      | 338.5 | 42   | 7    | 51    | USA (0.28)  |
| IV Prepared foodstuffs, beverages, tobacco | 253.8 | 37   | 10   | 53    | USA (0.19)  |
| XV Base metals                             | 238.4 | 41   | 3    | 56    | GBR (0.41)  |
| XIV Precious metals and stones             | 154.7 | 41   | 4    | 56    | CAN (0.62)  |
| I Live animals; animal products            | 131.1 | 34   | 14   | 51    | SAU (0.16)  |
| VI Chemicals                               | 121.8 | 49   | 14   | 37    | USA (0.22)  |
| XVII Transport equipment                   | 90.2  | 78   | 4    | 18    | DEU (0.30)  |
| XI Textiles and apparel                    | 74.9  | 49   | 3    | 48    | USA (0.52)  |
| III Fats and oils                          | 65.0  | 31   | 10   | 59    | CHE (0.24)  |
| XVI Machinery and electrical equipment     | 62.7  | 63   | 4    | 33    | USA (0.39)  |
| VII Plastics and rubber                    | 61.7  | 51   | 8    | 42    | USA (0.31)  |
| X Pulp, paper                              | 41.2  | 54   | 28   | 18    | PAN (0.53)  |
| XVIII Optical, precision instruments       | 32.7  | 72   | 8    | 19    | USA (0.61)  |
| IX Wood and articles                       | 23.6  | 27   | 34   | 39    | BHS (0.21)  |
| VIII Hides, skins, leather                 | 15.3  | 40   | 12   | 49    | BGR (0.15)  |
| XIII Stone, ceramics, glass                | 10.2  | 49   | 4    | 46    | USA (0.26)  |
| XX Miscellaneous manufactures              | 10.1  | 53   | 7    | 40    | USA (0.41)  |
| Unclassified                               | 6.7   | 49   | 29   | 23    | USA (0.31)  |
| XII Footwear, headgear                     | 4.6   | 48   | 1    | 50    | USA (0.58)  |
| XXI Works of art                           | 0.6   | 58   | 1    | 41    | BGR (0.50)  |
| XIX Arms                                   | 0.3   | 1    | 0    | 99    | CHE (0.51)  |
| Special / unclassified (98-99)             | 0.1   | 13   | 1    | 87    | USA (0.45)  |

HS sections ranked by export value. For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the section's export value, percent; Lead parent = largest parent country among the section's foreign MNEs and its share of the section's foreign-MNE value.

**Table 36:** Top 30 HS6 products by export value

| HS6    | Description                                                                                         | \$bn  | For. | Dom. | Local | Lead parent |
|--------|-----------------------------------------------------------------------------------------------------|-------|------|------|-------|-------------|
| 270900 | Petroleum oils and oils obt. from bituminous mins., crude                                           | 209.3 | 21   | 75   | 4     | USA (0.45)  |
| 260300 | Copper ores and concentrates                                                                        | 206.1 | 83   | 1    | 16    | GBR (0.54)  |
| 740311 | Cathodes and sections of cathodes, of refined copper, unwrought                                     | 137.9 | 32   | 3    | 65    | GBR (0.59)  |
| 710812 | Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)                           | 135.7 | 40   | 4    | 56    | CAN (0.66)  |
| 230400 | Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ...   | 97.5  | 35   | 12   | 52    | CHE (0.33)  |
| 270112 | Bituminous coal, whether/not pulverised but not agglomerated                                        | 68.1  | 77   | 0    | 23    | GBR (0.49)  |
| 120100 | Soya beans, whether/not broken                                                                      | 56.6  | 64   | 4    | 32    | USA (0.51)  |
| 100590 | Maize (corn), other than seed                                                                       | 43.1  | 53   | 7    | 40    | USA (0.37)  |
| 090111 | Coffee, not roasted, not decaffeinated                                                              | 42.4  | 34   | 14   | 52    | DEU (0.40)  |
| 271019 | Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...   | 40.9  | 21   | 50   | 30    | ESP (0.30)  |
| 150710 | Soya bean oil, crude, whether/not degummed, not chemically modified                                 | 38.7  | 33   | 13   | 53    | USA (0.26)  |
| 870421 | Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist...  | 29.1  | 85   | 0    | 15    | DEU (0.49)  |
| 271011 | Light petroleum oils and preparations                                                               | 21.4  | 18   | 6    | 76    | GBR (0.22)  |
| 020230 | Meat of bovine animals, frozen, boneless                                                            | 20.2  | 41   | 5    | 53    | SAU (0.48)  |
| 230120 | Flours, meals and pellets of fish/of crustaceans, molluscs/other aquatic invertebrates              | 17.8  | 31   | 10   | 58    | BMU (0.34)  |
| 740200 | Unrefined copper; copper anodes for electrolytic refining                                           | 17.7  | 76   | 1    | 23    | GBR (1.00)  |
| 100190 | Wheat other than durum wheat; meslin                                                                | 17.0  | 50   | 5    | 45    | USA (0.25)  |
| 220421 | Wine other than sparkling wine of fresh grapes, incl. fortified; grape must with fermentation ...   | 16.5  | 22   | 11   | 67    | FRA (0.15)  |
| 080390 | Bananas, fresh or dried (excl. plantains) [HS2012 code]                                             | 15.7  | 32   | 1    | 67    | JOR (0.34)  |
| 610910 | T-shirts, singlets and other vests, knitted/crocheted, of cotton                                    | 14.9  | 60   | 0    | 39    | USA (0.78)  |
| 260800 | Zinc ores and concentrates                                                                          | 14.7  | 72   | 0    | 27    | GBR (0.67)  |
| 080610 | Grapes, fresh                                                                                       | 13.3  | 29   | 9    | 62    | ESP (0.19)  |
| 260111 | Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated                           | 13.1  | 2    | 47   | 51    | CAN (0.64)  |
| 260700 | Lead ores and concentrates                                                                          | 12.9  | 58   | 0    | 42    | GBR (0.69)  |
| 870323 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s...  | 12.9  | 75   | 0    | 25    | FRA (0.45)  |
| 020130 | Meat of bovine animals, fresh/chilled, boneless                                                     | 12.7  | 35   | 11   | 55    | SAU (0.52)  |
| 901890 | Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electro-m...   | 12.2  | 89   | 0    | 11    | USA (0.48)  |
| 300490 | Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...   | 10.9  | 55   | 7    | 38    | USA (0.24)  |
| 382490 | Chemical products and preparations of the chemical/allied industries (including those consisting... | 10.8  | 36   | 14   | 50    | LIE (0.60)  |
| 470321 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, confi...   | 9.4   | 71   | 29   | 0     | PAN (0.82)  |

Top 30 HS6 products by export value (57% of the scope's exports). For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

**Table 37:** Top 30 HS6 products by foreign-MNE export value

| HS6    | Description                                                                                        | \$bn  | For. | Dom. | Local | Lead parent |
|--------|----------------------------------------------------------------------------------------------------|-------|------|------|-------|-------------|
| 260300 | Copper ores and concentrates                                                                       | 206.1 | 83   | 1    | 16    | GBR (0.54)  |
| 710812 | Gold (incl. gold plated with platinum), in unwrought forms (excl. powder)                          | 135.7 | 40   | 4    | 56    | CAN (0.66)  |
| 270112 | Bituminous coal, whether/not pulverised but not agglomerated                                       | 68.1  | 77   | 0    | 23    | GBR (0.49)  |
| 740311 | Cathodes and sections of cathodes, of refined copper, unwrought                                    | 137.9 | 32   | 3    | 65    | GBR (0.59)  |
| 270900 | Petroleum oils and oils obt. from bituminous mins., crude                                          | 209.3 | 21   | 75   | 4     | USA (0.45)  |
| 120100 | Soya beans, whether/not broken                                                                     | 56.6  | 64   | 4    | 32    | USA (0.51)  |
| 230400 | Oil-cake and other solid residues, whether/not ground/in pellets, from extraction of soyabean ...  | 97.5  | 35   | 12   | 52    | CHE (0.33)  |
| 870421 | Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist... | 29.1  | 85   | 0    | 15    | DEU (0.49)  |
| 100590 | Maize (corn), other than seed                                                                      | 43.1  | 53   | 7    | 40    | USA (0.37)  |
| 090111 | Coffee, not roasted, not decaffeinated                                                             | 42.4  | 34   | 14   | 52    | DEU (0.40)  |
| 740200 | Unrefined copper; copper anodes for electrolytic refining                                          | 17.7  | 76   | 1    | 23    | GBR (1.00)  |
| 150710 | Soya bean oil, crude, whether/not degummed, not chemically modified                                | 38.7  | 33   | 13   | 53    | USA (0.26)  |
| 901890 | Instruments and appliances used in medical/surgical/veterinary sciences, incl. other electrom...   | 12.2  | 89   | 0    | 11    | USA (0.48)  |
| 260800 | Zinc ores and concentrates                                                                         | 14.7  | 72   | 0    | 27    | GBR (0.67)  |
| 870323 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s... | 12.9  | 75   | 0    | 25    | FRA (0.45)  |
| 610910 | T-shirts, singlets and other vests, knitted/crocheted, of cotton                                   | 14.9  | 60   | 0    | 39    | USA (0.78)  |
| 271111 | Natural gas, liquefied                                                                             | 8.5   | 100  | 0    | 0     | USA (1.00)  |
| 100190 | Wheat other than durum wheat; meslin                                                               | 17.0  | 50   | 5    | 45    | USA (0.25)  |
| 720260 | Ferro-nickel, in granular/powder form                                                              | 8.5   | 100  | 0    | 0     | ZAF (0.85)  |
| 271019 | Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ...  | 40.9  | 21   | 50   | 30    | ESP (0.30)  |
| 020230 | Meat of bovine animals, frozen, boneless                                                           | 20.2  | 41   | 5    | 53    | SAU (0.48)  |
| 901839 | Catheters, cannulae and the like                                                                   | 9.2   | 89   | 0    | 11    | USA (0.86)  |
| 854231 | Electronic integrated circuits, processors and controllers, whether/not combined with memorie...   | 7.6   | 99   | 0    | 1     | USA (0.98)  |
| 260700 | Lead ores and concentrates                                                                         | 12.9  | 58   | 0    | 42    | GBR (0.69)  |
| 870322 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s... | 7.5   | 100  | 0    | 0     | USA (0.59)  |
| 470321 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, conif...  | 9.4   | 71   | 29   | 0     | PAN (0.82)  |
| 210690 | Food preparations, n.e.s.                                                                          | 7.8   | 81   | 1    | 18    | USA (0.77)  |
| 300490 | Medicaments (excluding goods of heading 30.02/30.05/30.06/3004.10-3004.50) consisting of mixed...  | 10.9  | 55   | 7    | 38    | USA (0.24)  |
| 870840 | Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.                    | 6.0   | 99   | 0    | 1     | DEU (0.74)  |
| 261610 | Silver ores and concentrates                                                                       | 7.6   | 76   | 0    | 24    | GBR (0.39)  |

Top 30 HS6 products by foreign-MNE export value (60% of foreign-MNE exports). For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

**Table 38:** HS6 products ( $\geq$  \$500m) with the highest foreign-MNE share

| HS6    | Description                                                                                        | \$bn | For. | Dom. | Local | Lead parent |
|--------|----------------------------------------------------------------------------------------------------|------|------|------|-------|-------------|
| 720260 | Ferro-nickel, in granular/powder form                                                              | 8.46 | 100  | 0    | 0     | ZAF (0.85)  |
| 790500 | Zinc plates, sheets, strip and foil                                                                | 0.51 | 100  | 0    | 0     | GBR (0.01)  |
| 790112 | Zinc, not alloyed, unwrought, containing by weight <99.99% of zinc                                 | 2.23 | 100  | 0    | 0     | BRA (1.00)  |
| 870321 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s... | 4.78 | 100  | 0    | 0     | NLD (0.49)  |
| 390410 | Poly(vinyl chloride), not mixed with any other substance, in primary forms                         | 4.37 | 100  | 0    | 0     | MEX (0.83)  |
| 271111 | Natural gas, liquefied                                                                             | 8.50 | 100  | 0    | 0     | USA (1.00)  |
| 870322 | Vehicles (excl. of 87.02 and 8703.10) principally designed for the transport of persons, with s... | 7.47 | 100  | 0    | 0     | USA (0.59)  |
| 870333 | Vehicles principally designed for the transport of persons (excl. of 87.02 and 8703.10-8703.24)... | 3.49 | 100  | 0    | 0     | JPN (1.00)  |
| 251010 | Natural calcium phosphates, natural aluminium calcium phosphates and phosphatic chalk, unground    | 2.78 | 100  | 0    | 0     | BRA (1.00)  |
| 280300 | Carbon (carbon blacks and other forms of carbon, n.e.s.)                                           | 0.81 | 100  | 0    | 0     | USA (0.99)  |
| 470311 | Chemical wood pulp, soda/sulphate, other than dissolving grades, unbleached, coniferous            | 1.90 | 100  | 0    | 0     | PAN (0.99)  |
| 300691 | Appliances identifiable for ostomy use                                                             | 1.55 | 100  | 0    | 0     | GBR (0.80)  |
| 853620 | Automatic circuit breakers, for a voltage not >1000V                                               | 3.61 | 99   | 0    | 1     | IRL (0.88)  |
| 854231 | Electronic integrated circuits, processors and controllers, whether/not combined with memorie...   | 7.62 | 99   | 0    | 1     | USA (0.98)  |
| 381512 | Supported catalysts, with precious metal/precious metal comps. as the active substance             | 0.73 | 99   | 0    | 1     | GBR (0.80)  |
| 870840 | Gear boxes and parts thereof, of the motor vehicles of headings 87.01 to 87.05.                    | 5.98 | 99   | 0    | 1     | DEU (0.74)  |
| 620711 | Men's/boys' underpants and briefs (excl. knitted/crocheted), of cotton                             | 0.53 | 98   | 0    | 2     | USA (0.93)  |
| 854430 | Ignition wiring sets and other wiring sets of a kind used in vehicles/aircraft/ships               | 2.24 | 98   | 0    | 2     | JPN (0.81)  |
| 740710 | Bars, rods and profiles, of refined copper                                                         | 0.63 | 98   | 0    | 1     | MEX (0.06)  |
| 390120 | Polyethylene having a sp.gr. of 0.94/more, in primary forms                                        | 1.16 | 98   | 0    | 2     | USA (0.98)  |
| 961900 | Sanitary towels and tampons, napkins and napkin liners for babies and similar sanitary article...  | 2.39 | 97   | 1    | 1     | USA (0.79)  |
| 350300 | Gelatin, incl. gelatin in rectangular (incl. square) sheets, whether/not surface-worked /colou...  | 0.78 | 97   | 0    | 3     | BEL (0.43)  |
| 790111 | Zinc, not alloyed, unwrought, containing by weight 99.99%/more of zinc                             | 3.48 | 97   | 0    | 3     | BRA (0.89)  |
| 521213 | Woven fabrics of cotton (excl. of 52.08-52.11), dyed, weighing not >200 g/m2                       | 1.04 | 96   | 0    | 4     | USA (1.00)  |
| 390190 | Polymers of ethylene, in primary forms (excl. of 3901.10-3901.30)                                  | 1.19 | 96   | 0    | 4     | USA (0.99)  |
| 840820 | Compression-ignition internal combustion piston engines (diesel/semi-diesel engines) of a kind...  | 1.76 | 96   | 0    | 4     | USA (0.54)  |
| 761010 | Doors, windows and their frames and thresholds for doors, of aluminium                             | 1.53 | 96   | 0    | 4     | CYM (0.98)  |
| 870422 | Motor vehicles for the transport of goods (excl. of 8704.10), with C-I internal combustion pist... | 0.95 | 96   | 0    | 4     | JPN (0.87)  |
| 281700 | Zinc oxide; zinc peroxide                                                                          | 0.74 | 95   | 0    | 5     | BRA (0.01)  |
| 300610 | Sterile surgical catgut, similar sterile suture materials (including sterile absorbable surgic...  | 1.12 | 95   | 3    | 2     | IRL (0.95)  |

HS6 products with export value  $\geq$  \$500m ranked by the foreign-MNE share (461 products qualify). For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

**Table 39: HS6 products ( $\geq$  \$500m) with the highest domestic-MNE share**

| HS6    | Description                                                                                       | \$bn  | For. | Dom. | Local | Lead parent |
|--------|---------------------------------------------------------------------------------------------------|-------|------|------|-------|-------------|
| 282520 | Lithium oxide and hydroxide                                                                       | 0.5   | 0    | 100  | 0     | –           |
| 481092 | Paper and paperboard(excl. of 4810.13-4810.39), coated on one/both sides with kaolin (China cl... | 2.2   | 1    | 99   | 0     | IRL (0.75)  |
| 310420 | Potassium chloride                                                                                | 2.0   | 3    | 95   | 2     | DZA (0.55)  |
| 441193 | Fibreboard of wood/other ligneous materials, whether/not bonded with resins/other organic subs... | 0.5   | 1    | 94   | 5     | ESP (0.94)  |
| 260112 | Iron ores and concentrates (excl. roasted iron pyrites), agglomerated                             | 1.8   | 7    | 88   | 4     | CHE (1.00)  |
| 350110 | Casein                                                                                            | 0.5   | 0    | 84   | 16    | CHL (0.59)  |
| 020727 | Cuts and edible offal of turkey, frozen                                                           | 0.5   | 1    | 82   | 17    | JPN (0.02)  |
| 730640 | Tubes, pipes and hollow profiles (excl. of 7306.10-7306.30), welded, of circular cross-section... | 0.5   | 15   | 82   | 3     | DEU (0.59)  |
| 270900 | Petroleum oils and oils obt. from bituminous mins., crude                                         | 209.3 | 21   | 75   | 4     | USA (0.45)  |
| 390390 | Polymers of styrene, in primary forms (excl. of 3903.11-3903.30)                                  | 0.5   | 15   | 73   | 12    | NLD (0.59)  |
| 440711 | Wood sawn/chipped length wise, sliced/peeled, whether/not planed, sanded/end-jointed, of a thi... | 2.5   | 10   | 71   | 19    | BHS (0.39)  |
| 441239 | Plywood, consisting solely of sheets of wood (other than bamboo,excl. of 4412.31 and4412.32), ... | 2.8   | 28   | 71   | 1     | VGB (0.45)  |
| 390319 | Polystyrene other than expansible, in primary forms                                               | 0.6   | 22   | 67   | 11    | PAN (0.99)  |
| 040610 | Fresh (unripened/uncured) cheese, incl. whey cheese, and curd                                     | 1.1   | 12   | 66   | 22    | MUS (0.34)  |
| 283691 | Lithium carbonates                                                                                | 7.1   | 14   | 65   | 21    | USA (0.54)  |
| 540219 | High tenacity yarn other than textured yarn/sewing thread, of nylon/other polyamides (excl. of... | 0.5   | 36   | 64   | 0     | GBR (0.01)  |
| 481141 | Gummed/adhesive paper,paperboard,coated, impregnated, covered, surface-coloured, surface-decor... | 0.8   | 26   | 62   | 12    | GBR (0.54)  |
| 902139 | Artificial parts of the body other than teeth, dental fittings and joints                         | 3.8   | 10   | 60   | 30    | VGB (0.45)  |
| 722830 | Bars and rods of other alloy steel (excl. of 72.27, 7228.10 and 7228.20), not further worked t... | 0.7   | 25   | 56   | 19    | SGP (0.59)  |
| 020329 | Meat of swine (excl. carcasses/half-carcasses/hams/shoulders and cuts thereof ), froxen           | 2.6   | 29   | 54   | 17    | JPN (0.55)  |
| 470329 | Chemical wood pulp, soda/sulphate, other than dissolving grades, semi-bleached/bleached, non-c... | 7.8   | 46   | 54   | 0     | PAN (1.00)  |
| 491110 | Trade advertising material, commercial catalogue-sandthe like                                     | 0.6   | 16   | 53   | 31    | USA (0.25)  |
| 880240 | Aeroplanes and other aircraft, of an unladen weight >15000kg                                      | 5.4   | 40   | 53   | 7     | CHL (0.99)  |
| 410449 | Tanned/crust hides and skins of bovine (incl. buffalo)/equine animals, without hair on, in the... | 0.7   | 37   | 52   | 12    | URY (0.67)  |
| 271019 | Petroleum oils and oils obtained from bituminous minerals (other than crude) and preparations ... | 40.9  | 21   | 50   | 30    | ESP (0.30)  |
| 210111 | Extracts, essences and concentrates of coffee                                                     | 2.9   | 23   | 48   | 29    | CHE (0.28)  |
| 260111 | Iron ores and concentrates (excl. roasted iron pyrites), non-agglomerated                         | 13.1  | 2    | 47   | 51    | CAN (0.64)  |
| 330300 | Perfumes and toilet waters                                                                        | 1.5   | 34   | 46   | 20    | PAN (0.36)  |
| 030319 | Pacific salmon (Oncorhynchus gorbuscha/keta/tshawytscha/kisutch/masou/rhodurus), frozen (excl...  | 0.7   | 8    | 44   | 47    | CAN (0.39)  |
| 391990 | Self-adhesive plates, sheets, film, foil, tape, strip and other flat shapes, of plastics, whet... | 0.8   | 41   | 44   | 15    | GBR (0.50)  |

HS6 products with export value  $\geq$  \$500m ranked by the domestic-MNE share. For./Dom./Local = foreign-MNE, domestic-MNE and local-firm shares of the product's export value, percent (pooled 2006–2022, nine LAC origins). Foreign = matched firms whose parent is abroad or unknown; Domestic = parent in the exporting country; Local = unmatched. Lead parent = largest parent country among the product's foreign MNEs and its share of the product's foreign-MNE value.

**Table 40:** Distribution of the HS6 foreign share, numbers

| Foreign share bin (%) | Share of value | MNE share of exporting firms | N HS6 |
|-----------------------|----------------|------------------------------|-------|
| 0–10                  | 0.048          | 0.250                        | 1,426 |
| 10–20                 | 0.054          | 0.285                        | 731   |
| 20–30                 | 0.189          | 0.329                        | 652   |
| 30–40                 | 0.202          | 0.372                        | 591   |
| 40–50                 | 0.127          | 0.411                        | 519   |
| 50–60                 | 0.059          | 0.433                        | 442   |
| 60–70                 | 0.070          | 0.435                        | 386   |
| 70–80                 | 0.074          | 0.470                        | 311   |
| 80–90                 | 0.129          | 0.479                        | 311   |
| 90–100                | 0.048          | 0.530                        | 560   |

MNE share of exporting firms = matched firm-cells (firm x destination x HS6 x year) over all firm-cells in the bin's products.

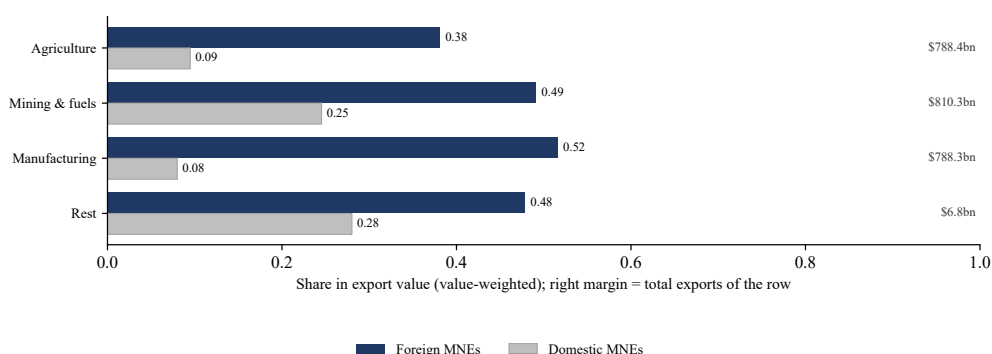
## 9 The four sectors at a glance (item 2)

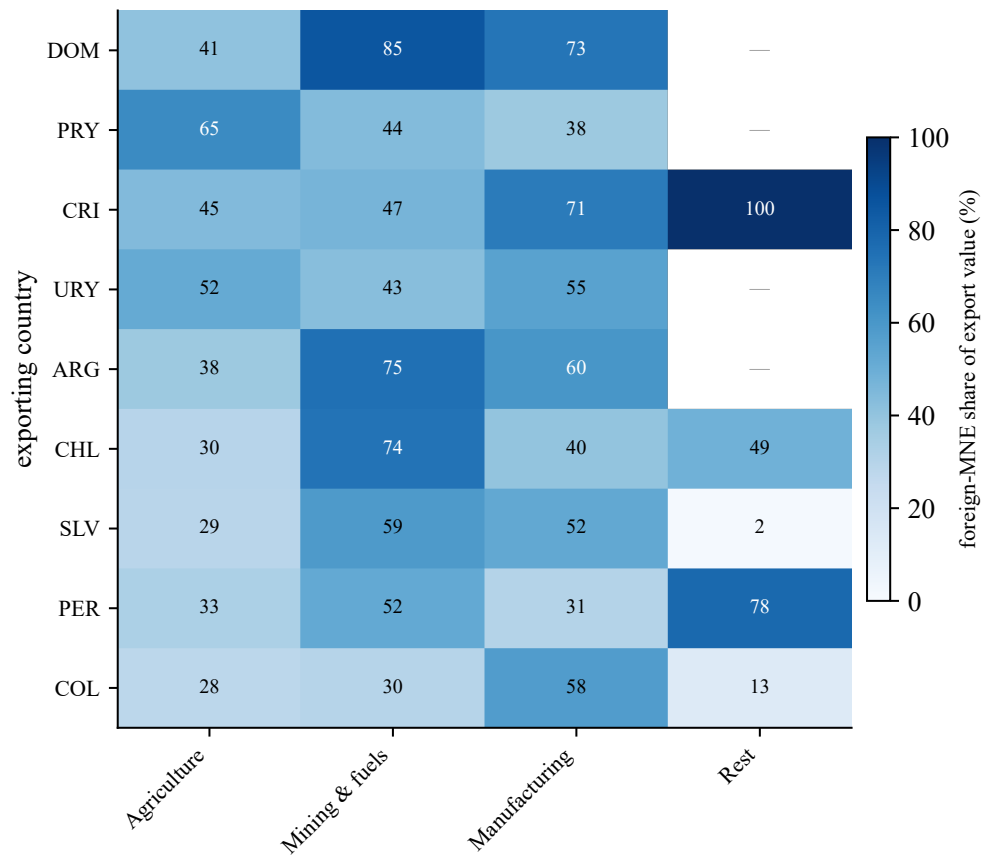
Sector definitions in the conventions above. The full set of exhibits within each sector is in the companion document *WP\_sectors*.

**Table 41:** The four sectors: total and MNE exports, shares, leading parents

| Sector         | Total exports (\$bn) | MNE exports (\$bn) | Foreign (%) | Domestic (%) | Leading parents (share of foreign) |
|----------------|----------------------|--------------------|-------------|--------------|------------------------------------|
| Agriculture    | 788.4                | 375.3              | 38.1        | 9.5          | USA 0.24, CHE 0.11, NLD 0.10       |
| Mining & fuels | 810.3                | 597.2              | 49.2        | 24.5         | GBR 0.36, USA 0.21, CAN 0.16       |
| Manufacturing  | 788.3                | 470.1              | 51.6        | 8.0          | USA 0.25, GBR 0.14, DEU 0.09       |
| Rest           | 6.8                  | 5.2                | 47.9        | 28.0         | USA 0.39, GBR 0.12, ESP 0.11       |
| All            | 2,393.8              | 1,447.8            | 46.3        | 14.1         |                                    |

Agriculture = HS 01–24; Mining & fuels = HS 25–27 and 71 (precious metals and stones); Manufacturing = HS 28–97 (excl. 71); Rest = HS 98–99, chapter 00 and unclassifiable codes. Services are not observable in customs merchandise data. Pooled 2006–2022, nine LAC origins. Foreign / Domestic = share of the row's export value; leading parents = the three largest parent countries among the row's foreign MNEs with their share of the row's foreign-MNE value.

**Figure 30:** Foreign and domestic MNE shares by sector.



**Figure 31:** Foreign-MNE share by origin and sector (%).

**Table 42:** Composition (Panel A) and foreign-MNE share (Panel B) by origin and sector

| Origin                                                                                | Agriculture | Mining & fuels | Manufacturing | Rest  | All   |
|---------------------------------------------------------------------------------------|-------------|----------------|---------------|-------|-------|
| <i>Panel A: composition of the origin's exports in the scope, % (rows sum to 100)</i> |             |                |               |       |       |
| ARG                                                                                   | 57.4        | 9.9            | 32.7          | 0.0   | 100.0 |
| CHL                                                                                   | 21.9        | 29.7           | 46.9          | 1.4   | 100.0 |
| COL                                                                                   | 16.3        | 63.4           | 20.3          | 0.0   | 100.0 |
| CRI                                                                                   | 41.5        | 1.0            | 57.5          | 0.0   | 100.0 |
| DOM                                                                                   | 23.1        | 22.8           | 54.1          | 0.0   | 100.0 |
| PER                                                                                   | 19.4        | 61.0           | 19.6          | 0.0   | 100.0 |
| PRY                                                                                   | 85.2        | 1.1            | 13.7          | 0.0   | 100.0 |
| SLV                                                                                   | 22.1        | 3.0            | 74.9          | 0.0   | 100.0 |
| URY                                                                                   | 63.0        | 2.4            | 34.6          | 0.0   | 100.0 |
| All origins                                                                           | 32.9        | 33.9           | 32.9          | 0.3   | 100.0 |
| <i>Panel B: foreign-MNE share of export value within the cell, %</i>                  |             |                |               |       |       |
| ARG                                                                                   | 37.8        | 75.1           | 60.4          | —     | 48.9  |
| CHL                                                                                   | 29.6        | 73.9           | 39.9          | 48.6  | 47.9  |
| COL                                                                                   | 28.0        | 30.1           | 57.8          | 12.7  | 35.3  |
| CRI                                                                                   | 44.5        | 46.9           | 70.9          | 100.0 | 59.7  |
| DOM                                                                                   | 40.7        | 85.3           | 73.2          | —     | 68.4  |
| PER                                                                                   | 33.0        | 52.2           | 30.6          | 78.0  | 44.2  |
| PRY                                                                                   | 65.0        | 44.1           | 37.9          | —     | 61.1  |
| SLV                                                                                   | 28.9        | 58.8           | 52.5          | 1.5   | 47.4  |
| URY                                                                                   | 51.8        | 42.9           | 55.3          | —     | 52.8  |
| All origins                                                                           | 38.1        | 49.2           | 51.6          | 47.9  | 46.3  |

Pooled 2006–2022. Panel A: share of each sub-sector in the origin's exports within the scope (rows sum to 100). Panel B: foreign-MNE share of export value in each origin × sub-sector cell ('—' = no exports); 'All' = the origin's overall foreign-MNE share in the scope.

**Table 43:** Foreign-MNE exports: destination region by sector (row %)

| Sector / Destination region | Latin America | North America | Europe | Asia | Africa | Oceania | Rest of World | Total |
|-----------------------------|---------------|---------------|--------|------|--------|---------|---------------|-------|
| Agriculture                 | 31.1          | 13.8          | 20.2   | 28.1 | 6.1    | 0.4     | 0.4           | 300.6 |
| Mining & fuels              | 14.3          | 14.6          | 24.7   | 45.5 | 0.5    | 0.2     | 0.2           | 398.3 |
| Manufacturing               | 50.8          | 20.7          | 9.3    | 17.3 | 0.7    | 1.1     | 0.2           | 407.0 |
| Rest                        | 37.3          | 42.2          | 17.2   | 2.9  | 0.1    | 0.2     | 0.1           | 3.3   |

Foreign-MNE exports: destination mix of each sector, row percentages; last column = row total, USD bn.

**Table 44:** Memo: service-sector affiliates among matched exporters (NAICS of the affiliate)

| NAICS sector of the matched exporting affiliate      | % of matched export value |
|------------------------------------------------------|---------------------------|
| goods-producing or unknown                           | 82.4                      |
| information, finance, professional, holdings (51–56) | 7.7                       |
| wholesale / retail trade (42–45)                     | 7.4                       |
| other services (61–92)                               | 1.8                       |
| transport, logistics (48–49)                         | 0.6                       |

Memo. Customs data record goods; ‘services’ can only be approached through the affiliate’s own industry code. Matched exporters whose NAICS is a trade, logistics, finance or holding code export goods produced elsewhere in the group or bought from local suppliers.

## A Numbers behind the figures

### A.1 All goods

**Table 45:** Figure 1 numbers (foreign, domestic, total)

| Origin | Foreign | Domestic | MNE total | Value (\$bn) |
|--------|---------|----------|-----------|--------------|
| COL    | 0.353   | 0.391    | 0.744     | 528.0        |
| DOM    | 0.684   | 0.013    | 0.697     | 68.8         |
| CRI    | 0.597   | 0.036    | 0.633     | 104.7        |
| URY    | 0.528   | 0.100    | 0.628     | 70.3         |
| PRY    | 0.611   | 0.011    | 0.622     | 65.4         |
| CHL    | 0.479   | 0.136    | 0.615     | 478.7        |
| ARG    | 0.489   | 0.075    | 0.564     | 597.9        |
| SLV    | 0.474   | 0.003    | 0.477     | 65.4         |
| PER    | 0.442   | 0.023    | 0.466     | 414.5        |

**Table 46:** Figure 1 by parent, numbers

| Origin | USA   | GBR   | CAN   | LIE   | DEU   | NLD   | CHE   | JPN   | PAN   | BRA   | Other | Dom.  | MNE total | Value (\$bn) |
|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|--------------|
| COL    | 0.116 | 0.052 | 0.012 | 0.000 | 0.013 | 0.002 | 0.002 | 0.004 | 0.010 | 0.004 | 0.139 | 0.391 | 0.744     | 528.0        |
| DOM    | 0.170 | 0.030 | 0.161 | 0.003 | 0.041 | 0.000 | 0.004 | 0.007 | 0.005 | 0.015 | 0.249 | 0.013 | 0.697     | 68.8         |
| CRI    | 0.303 | 0.021 | 0.001 | 0.000 | 0.025 | 0.015 | 0.019 | 0.023 | 0.023 | 0.000 | 0.166 | 0.036 | 0.633     | 104.7        |
| URY    | 0.049 | 0.018 | 0.009 | 0.000 | 0.010 | 0.038 | 0.009 | 0.024 | 0.007 | 0.085 | 0.277 | 0.100 | 0.628     | 70.3         |
| PRY    | 0.301 | 0.003 | 0.000 | 0.000 | 0.000 | 0.021 | 0.000 | 0.016 | 0.001 | 0.009 | 0.259 | 0.011 | 0.622     | 65.4         |
| CHL    | 0.026 | 0.210 | 0.070 | 0.062 | 0.005 | 0.000 | 0.004 | 0.011 | 0.024 | 0.001 | 0.067 | 0.136 | 0.615     | 478.7        |
| ARG    | 0.081 | 0.029 | 0.023 | 0.039 | 0.045 | 0.055 | 0.048 | 0.034 | 0.000 | 0.013 | 0.121 | 0.075 | 0.564     | 597.9        |
| SLV    | 0.279 | 0.014 | 0.011 | 0.000 | 0.005 | 0.005 | 0.000 | 0.014 | 0.038 | 0.002 | 0.106 | 0.003 | 0.477     | 65.4         |
| PER    | 0.078 | 0.112 | 0.035 | 0.004 | 0.002 | 0.001 | 0.005 | 0.002 | 0.028 | 0.026 | 0.148 | 0.023 | 0.466     | 414.5        |

Figure 1 split by parent country. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and ‘Other foreign’ (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country.

**Table 47:** Figure 2 by parent, numbers

| PCI quintile | USA   | GBR   | CAN   | LIE   | DEU   | NLD   | CHE   | JPN   | PAN   | BRA   | Other | Dom.  | MNE total | Value (\$bn) |
|--------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|--------------|
| Q1           | 0.111 | 0.130 | 0.058 | 0.032 | 0.002 | 0.010 | 0.007 | 0.001 | 0.008 | 0.008 | 0.093 | 0.181 | 0.641     | 1,176.3      |
| Q2           | 0.072 | 0.074 | 0.014 | 0.031 | 0.006 | 0.027 | 0.046 | 0.008 | 0.019 | 0.021 | 0.125 | 0.098 | 0.540     | 450.3        |
| Q3           | 0.183 | 0.027 | 0.005 | 0.000 | 0.042 | 0.009 | 0.004 | 0.017 | 0.069 | 0.017 | 0.186 | 0.129 | 0.689     | 147.6        |
| Q4           | 0.188 | 0.041 | 0.013 | 0.004 | 0.040 | 0.013 | 0.015 | 0.014 | 0.010 | 0.007 | 0.254 | 0.087 | 0.686     | 129.6        |
| Q5           | 0.088 | 0.008 | 0.005 | 0.017 | 0.153 | 0.072 | 0.019 | 0.146 | 0.002 | 0.007 | 0.137 | 0.071 | 0.724     | 136.7        |

Figure 2 split by parent country. Quintiles of the Hausmann–Hidalgo Product Complexity Index over HS6 products. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and ‘Other foreign’ (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country.

**Table 48:** Figure 3 by parent, numbers

| Technology category        | USA   | GBR   | CAN   | LIE   | DEU   | NLD   | CHE   | JPN   | PAN   | BRA   | Other | Dom.  | MNE total | Value (\$bn) |
|----------------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-----------|--------------|
| High tech manufacturing    | 0.220 | 0.012 | 0.003 | 0.000 | 0.034 | 0.005 | 0.023 | 0.011 | 0.006 | 0.009 | 0.210 | 0.080 | 0.613     | 60.4         |
| Medium tech manufacturing  | 0.097 | 0.018 | 0.006 | 0.022 | 0.072 | 0.036 | 0.029 | 0.049 | 0.006 | 0.009 | 0.164 | 0.081 | 0.587     | 492.5        |
| Low tech manufacturing     | 0.116 | 0.011 | 0.127 | 0.001 | 0.006 | 0.006 | 0.005 | 0.005 | 0.016 | 0.008 | 0.140 | 0.059 | 0.501     | 307.7        |
| Primary and resource-based | 0.094 | 0.128 | 0.025 | 0.030 | 0.003 | 0.013 | 0.013 | 0.004 | 0.018 | 0.014 | 0.114 | 0.182 | 0.638     | 1,436.4      |

Figure 3 split by parent country. Lall (2000) technology classification, four categories. Value-weighted; pooled 2006–2022; nine LAC origins (Ecuador excluded). The foreign-MNE bar is split into the 10 largest parent countries by foreign-MNE export value in the scope and ‘Other foreign’ (all remaining parents, including matched firms with no recorded parent country); domestic MNEs have a parent in the exporting country.